

	Department	Date	Signature
Drawn	EBS4-MS	21.07.2006	EBS4-MS
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Approved	EBS4 – Andrew Peter	21.07.2006	EBS4

1. ALTERATION LIST

Issue	ECR Number	Alteration	By	Valid From	Checked
7		4.17.4 Personalisation setting limitations/conditions -Added. 4.18 Network Management (MS-CAN) -Timing diagrams removed.	EBS4-MS	21.11.2006	
6		Sec 4.7.3 Mood Lighting New requirement with mood lighting Sec 4.8.3 Secure Tailgate Strategy (For Orion Only) Update spec. Sec 4.17 Personalisation Interface Updated to support CAN message (\$315)	EBS4-MS	25.06.2006	PJM-SW
5		Sec 4.1: Add Trailer Sense input and Rear Seat Belt input details. Sec 4.19 Power Management Sec 4.16 General Alarm Description Updated. 4.12.1.3 Diagnosis The way to detect errors is modified to support the circuit.	EBS4-MS	10.05.2006	
4	-	Following sections are added or modified after DTM 31st March 2006: Sec 4.7.3: Mood Lighting Dome lamp will ramp up to 100%. Configurable ramp up rate. Sec 4.8.10: Taxi Mode State diagram and description modified to support central lock and central unlock commands Sec 4.8.3: Secure Tailgate and Rear Window Strategy Add: After a BEM reset assume BEM in secure boot mode, except in plant mode. Sec 4.10: Update CAN message for car door locked state. Sec 4.12: Electronic Immobilisation System If number of DST keys trained is read as more than 8, then assume that BEM is in Plant Mode and number of DST keys trained is 0 (update to eeprom). Sec 4.15 Personalisation Interface Each personalisable function have additional option of 'inhibited'. Add default settings for personalisable functions. Sec 4.17.4 Wake-Up Events for MS-CAN by BCM -Sleep after 5 seconds unless other conditions hold the CAN alive. -Add a table for CAN bit description. -Change column description Signal to Signal/Function. Sec 4.11.3.1 Headlamp Chime Warning -Clear Headlamp Chime Warning bit on ignition switch in 'OFF' position.	EBS4-MS	07.04.2006	
3	-	Following sections are added or modified: Sec 4.7.2 Mood Lighting New function. Sec 4.11 Automatic Headlamps Table for Day/Night CAN bit setting. Sec 4.13 One-Touch Start New Requirement. Sec 4.14 One Touch Start New Requirement. Sec 4.15 General Alarm Description New Requirement. SDS Reviewed. Sec 4.18 Battery Over Voltage Strategies Sec 4.19 Enabling/Disabling DTC Logging	EBS4-MS	28.03.2006	

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Issue	ECR Number	Alteration	By	Valid From	Checked
2	-	Following sections are added or modified: Sect 4.3 - Front Wiper and Washer System Park on ignition off, Controlling high and low wipes. Wipe after wash. Sect 4.4 Rear Wiper and Washer System Park on ignition off. Wipe after wash. Sect 4.8.2 Manual (and Remote) Tailgate and Rear Window Release Tailgate button on the tailgate introduced for Orion and switch is enabled/disabled like Territory Sect 4.8.3 Secure Tailgate and Rear Window Strategy New function. Sect 4.8.4 Manual Central Locking Functions Central Lock switch and Central Unlock sw functions introduced. Sect 4.8.8 Drive-Away Locking New function Sect 4.11 Automatic Headlamps Modified to support new hardware. Sect 4.14 Personalisation Interface New Requirement Sect 4.15 Network Management (MS-CAN) New Requirement Sect 4.16 General Alarm Description New function. Sect 4.18 Configurable Options Sect 4.19 Timing Parameters	EBS4-MS	15.03.2006	
1	-	Initial draft is based on Document "F005VP0376 E265 RBEM Engineering Specification.doc – Issue 6.0 Dated 01-10-2003" and decision taken during Design Team Meeting conducted on 29-07-2005.	EBS1-MS	08.08.2005	

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3. General

3.1 Definitions

The following definitions shall apply throughout this specification:

- N/A Not applicable.
- VIL Voltage input low state.
- VIH Voltage input high state.
- VOL Voltage output low state (for solid state outputs).
- VOH Voltage output high state (for solid state outputs).
- VOUS Voltage output unenergised state (for switched outputs).
- VOES Voltage output energised state (for switched outputs).
- ∇ Critical Characteristic (Ford)
- SC Significant Characteristic (Ford)
- SDS System Design Specification (Ford)
- DS Design Specification (Ford)
- ES Engineering Specification (Ford)
- MRDB Master Reference Database (Ford)
- WDR Worldwide Design Standard (Ford)
- FPDS Ford Product Development System
- SWS Sumitomo Wiring Systems

3.2 Part and Document References

Description	
RBAU sales drawing	
RBAU Ford part numbers	
RBAU vehicle wiring diagram	
RBAU product brief	
RBAU system FMEA	
RBAU design FMEA	
RBAU control plan	
RBAU Orion Body Electronic Module sales drawing	

3.3 Functional Overview

Symbols in bold define the functional class for each function.

Class	Functionality
C*	Transponder challenge response immobilisation (* B1 for EMC Testing)
A	Ford Shark Protocol RF receiver interface
A	Remote central locking with selectable one or two stage unlock and hazard acknowledgment (Hazard acknowledgements and warnings are signalled via CAN)
A	Remote control for illuminated entry
A	Remote tailgate or rear window external switch release enabled with hazard acknowledgment
A	Door unlocked horn and hazard warning
A	DST Key in lock horn and hazard warning
A	Remote panic alarm using horn and hazards
A	Automatic door re-lock timer for remote entry
B	Manual control for dome lamp, illuminated entry delay and illuminated exit delay
B	Manual central locking control
B	Manual tailgate and rear window release with security time out and speed dependant disable
A	Priority key signalling via CAN
A	Door ajar and boot ajar signalling via CAN
B	Battery saver for dome, boot, glove box lamps and Audio One Hour Radio Play
A	Automatic dome lamp illumination triggered by ignition DST key removal or ignition off with software enable facility
B	Heated backlight (rear demist) with automatic time out
B	Front intermittent wiper dwell timer
B	Rear intermittent wiper dwell timer with reverse on
A	Either variable dwell with speed dependency, or fixed dwell; selected by automatic hardware detection
B	Front wipe after wash timer with fixed number of wipes
A	Rear wipe after wash timer with fixed number of wipes
A	Low noise relay for wiper functions
B	Power window supply control
A	Accessories line status via CAN
B1	125kb/s CAN interface
B	Ford diagnostic interface via CAN
C	Automatic headlamp control with twilight sensing, delayed turn-off and failsafe strategies with Headlamp On/Off threshold change due to wiper operation
A	Front washer motor control with minimum wash time and wipe after wash
A	Sun-load sensor interface and signal via CAN
A	Memory seat interface for priority key information

4. FUNCTIONAL DESCRIPTION

4.1 BEM Input Status

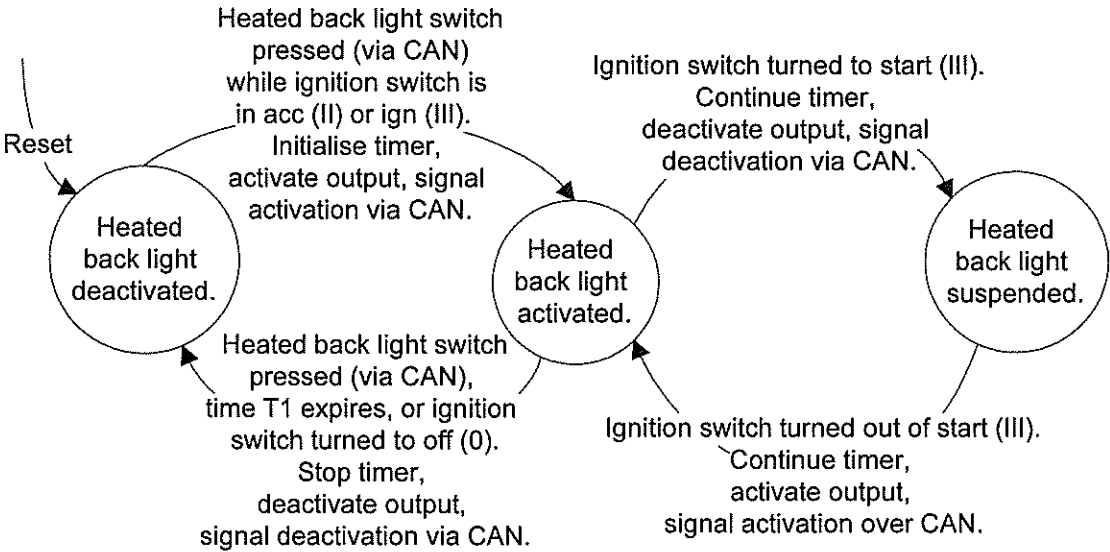
The BEM reports the status of the following inputs in the CAN bus:

Message	Signal Name	Possible Values	Comments
CAN Msg \$403	RFD_Ajar	No (0) Yes (1)	
	LFD_Ajar	No (0) Yes (1)	
	RRD_Ajar	No (0) Yes (1)	
	LRD_Ajar	No (0) Yes (1)	
	BootTailgateAjar	No (0) Yes (1)	
	AccessoriesSwitch	Off or Position 3 Position 1 or Position 2	
	FrontWiperIntermittent	No (0) Yes (1)	
	BonnetAjar	No (0) Yes (1)	
	RFD_Locked	No (0) Yes (1)	
CAN Msg \$406	Ignition_Status	Unknown Off Accessory (Pos1) Run (Pos2) Start (Pos3) Invalid	
	SeatbeltStatusRearLeft	Unlatched (0) Latched (1)	When input changes, wake CAN and send CAN message.
	SeatbeltStatusRearCentre	Unlatched (0) Latched (1)	
	SeatbeltStatusRearRight	Unlatched (0) Latched (1)	
	Trailer Sense Input		Notify on CAN when ignition is ON.

4.2 Heated Backlight Timer

The heated backlight relay may be activated (VOL) with the ignition switch in the accessories or ignition position. Activation is achieved when the pressing of the heated backlight switch is reported via the CAN bus. The relay output will deactivate (VOH) when time T01 elapses, the heated backlight switch is reported as being pressed again, or the ignition switch is turned to the off position. While the DST ignition key is being held in the start position, the heated backlight relay is temporarily deactivated, and will resume the previous state when the DST ignition key is released from start.

The operational status of the heated backlight is indicated via the CAN bus.



State diagram for heated backlight control
For any discrepancies, the functional description supersedes the above state diagram

If the rear demist switch is pressed and held, while the Ignition switch is turned from Off to Accessories position, the heated backlite relay turns on.

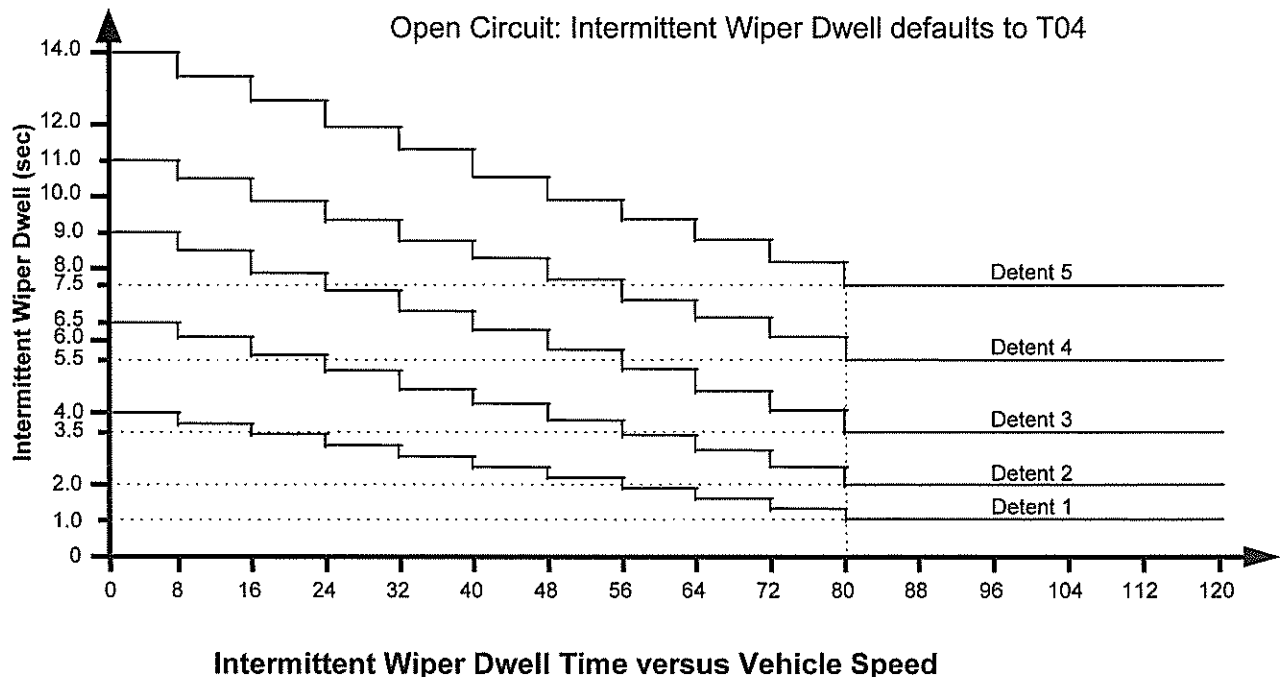
If rear demist switch is pressed and held while the Ignition switch is released from Start to Ignition, the heated backlite relay will change state.

Activation of the Heated Backlite is enabled only when the battery voltage is above 10.7 volts and disabled when battery voltage is below 10.3 volts.

4.3 Front Wiper and Washer System

4.3.1 Front Intermittent Wiper Timer with Variable Dwell, Speed Dependency, and Wipe on Take-off

Switching the front intermittent wiper switch to the intermittent position while the ignition switch is in the accessories or ignition position will enable the intermittent wipe function. While enabled, there will be a single wipe followed by a delay before the next wipe is initiated. If the wiper dwell potentiometer is disconnected (open), then the delay between wipes shall be of a fixed duration (T04 @ V0US). If, however, the vehicle is fitted with a wiper dwell potentiometer, then the delay (T03 @ V0US) is dependent on the potentiometer position. With the ignition switch in the ignition position, the dwell time is also affected by vehicle speed, reducing as speed increases, as per the figure and table below. When the vehicle accelerates to 8 km/h (as reported via CAN), with the wiper switch in the intermittent position, and the ignition switch in the ignition position, then the next wipe will be brought forward to implement a wipe on take-off function. It shall occur a short time after the previous wipe (T08), or immediately if this time has already elapsed. If the wiper travel time is longer than the time expected such as in the case where the Detent is in position 1, travelling at 80km/h, the wiper park switch feedback is monitored, allowing the next wipe to occur immediately with minimal delay. A timeout has been implemented just in case Wiper Park switch feedback fails. When no vehicle speed data is sent via CAN, the BEM will assume 0 km/h. The BEM's circuitry allows the wipers to driven to park position at the end of each wipe.



Intermittent Wiper Dwell Time versus Vehicle Speed

Detent Position	Short to GND	1	2	3	4	5	Open Circuit
Speed (km/h)	Front Intermittent Wiper Off Time (seconds)						
0	T03 = 4.1	T03 = 4.1	T03 = 6.6	T03 = 9.1	T03 = 11.1	T03 = 14.1	T04 = 7.0
8	T03 = 3.8	T03 = 3.8	T03 = 6.2	T03 = 8.6	T03 = 10.6	T03 = 13.5	T04 = 7.0
16	T03 = 3.5	T03 = 3.5	T03 = 5.7	T03 = 8.0	T03 = 10.0	T03 = 12.8	T04 = 7.0
24	T03 = 3.2	T03 = 3.2	T03 = 5.3	T03 = 7.5	T03 = 9.5	T03 = 12.2	T04 = 7.0
32	T03 = 2.9	T03 = 2.9	T03 = 4.8	T03 = 6.9	T03 = 8.9	T03 = 11.5	T04 = 7.0
40	T03 = 2.6	T03 = 2.6	T03 = 4.4	T03 = 6.4	T03 = 8.4	T03 = 10.9	T04 = 7.0
48	T03 = 2.3	T03 = 2.3	T03 = 3.9	T03 = 5.8	T03 = 7.8	T03 = 10.2	T04 = 7.0
56	T03 = 2.0	T03 = 2.0	T03 = 3.5	T03 = 5.3	T03 = 7.3	T03 = 9.6	T04 = 7.0
64	T03 = 1.7	T03 = 1.7	T03 = 3.0	T03 = 4.7	T03 = 6.7	T03 = 8.9	T04 = 7.0
72	T03 = 1.4	T03 = 1.4	T03 = 2.6	T03 = 4.2	T03 = 6.2	T03 = 8.3	T04 = 7.0
80 +	T03 = 1.1	T03 = 1.1	T03 = 2.1	T03 = 3.6	T03 = 5.6	T03 = 7.6	T04 = 7.0

Note: Since the mechanical design of the lever combination switch has detents 4 and 5 next to each other, there is a risk that detents 4 and 5 will not be distinguishable in production e.g. detent 5 may always appear as detent 4.

4.3.2 Front Washer Motor Control with Wipe after Wash

The front washer motor is activated (VOES) by pressing the front wash switch (momentary) to the wash position, with the ignition switch in the accessories or ignition position. The line is held by the BEM in the activated state for T05. The operator can obtain wash for longer than this time by continuing to hold the wash switch. After the washer motor has been running for T06, the wipers will automatically operate (VOES), and will continue to do so for the period that the washers are active. When the washers turn off, the wipers will perform three wipes, including the one already in progress, by monitoring the feedback from the wiper park switch and then stop (VOUS). In the event of failure of the feedback, the wipers will stop after a time approximately equivalent to three wipes (T07) following deactivation of the washer motor. Switching the ignition switch to the off or start positions will abort any current wash function. Operation of the intermittent wipe function is suspended while a wash function is in progress.

4.3.3 Drip Wipe

When front wiper switch is in 'OFF' position, after a wipe after wash and a delay of T81, there will be a drip wipe (one wipe). Drip wipe can be enabled or disabled by diagnostics.

4.3.4 Front Wiper Redundant Circuit Failure Notification

TBA

4.4 Rear Wiper and Washer System

4.4.1 Rear wiper motor control

Operation of rear wiper is depending on the front wiper switch position, rear wiper switch position and gear position.

The rear wiper can be in one of the following modes:

- Wiper Off
- Wiper Fully ON (Battery voltage is constantly supplied to the rear wiper motor)
- Wiper Operating with intermittent dwell time of T64. Dwell time T64 can be programmed at BOSCH End of Line Testing. If the value programmed is not in between 3 seconds and 14 seconds, a default value of 4 seconds will be used.

Rear Wiper will be OFF for the following conditions:

- Both front wiper switch and rear wiper switch in OFF position.
- Rear wiper switch in OFF position, and gear not in reverse position.

Rear Wiper will be fully ON for the following conditions:

- Rear wiper switch in ON position, and gear in reverse position.
- Front wiper switch in position I or II, and gear in reverse position.

Rear Wiper will be operating intermittently for the following conditions:

- Rear wiper switch in ON position and gear is not in reverse position.
- Front wiper switch in intermittent position, rear wiper switch in OFF position, and gear is in reverse position.

Rear wiper operation will be disabled when rear window or tailgate is ajar except in Plant Mode.

The following table summarises the above conditions.

Front Wiper Switch Position	Rear Wiper Switch Position	Gear in Reverse (via CAN)	Rear Wiper Motor
Off	Off	Yes	Off
Off	On	Yes	Continuous On
Off	On	No	*T64 sec Intermittent delay
Intermittent	Off	Yes	*T64 sec Intermittent delay
Intermittent	Off	No	Off
Intermittent	On	Yes	Continuous On
Intermittent	On	No	*T64 sec Intermittent delay
Low or High Speed (I or II)	Off	Yes	Continuous On
Low or High Speed (I or II)	Off	No	Off
Low or High Speed (I or II)	On	Yes	Continuous On
Low or High Speed (I or II)	On	No	*T64 sec Intermittent delay

Note: * - EEPROM memory programmable at RBAU Plant

4.4.1.1 Detecting Gear in Reverse position

Reverse Gear position can be determined by reading TransGearPos byte of CAN Msg \$330. This is assumed valid when the ignition switch in Ignition position. Rear wiper

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Operation due to reverse gear will stop when the Ignition switch is in the Accessories position. While the front wiper is operating and rear wiper switch not selected, if the reverse gear is selected, there will be a delay of T77 before the rear wiper will operate.

4.4.1.2 Rear Wiper Operation after Rear Tailgate/Window ajar

Rear wiper operation will be disabled when rear window or tailgate is ajar except when the BEM is in Plant Mode (Can be enabled or disabled by Bosch during production). If rear tailgate/window is closed, then there will a delay of T22 before rear wiper operation is enabled.

4.4.2 Rear Wipe after Wash Control

Rear Wipe after Wash function can be invoked by momentarily pressing the rear wash switch to the wash position, with the ignition switch in the accessories or ignition position. After the washer motor has been running for T63, the wipers will automatically operate (V_{OES}), and will continue to do so for the period that the washers are on. When the washer is turned off, the rear wiper will continue to operate for another 3 wipes. Switching the ignition switch to the off or start positions will abort any current wash and/or wipe function. Operation of the intermittent wipe function is suspended (wipers to have continuous wipe) while a wash function is in progress. Rear wiper operation will be disabled when rear window or tailgate is ajar. Rear Wiper motor is activated for a maximum of T67 as a result of wash function being invoked. This is to protect against scratching of windscreen due to rear washer switch failure.

If rear wash switch is pressed and released while rear tailgate/window ajar, then after rear tailgate/window closed and time delay T22, rear wipe operates for T68. If rear wipe after wash function is interrupted by rear tailgate/window ajar, then after rear tailgate/window closed and time delay T22, rear wipe operates for additional 3 wipes. Turning ignition switch to off position will cancel rear wipe after wash function.

4.4.3 Front wiper status on CAN

Two CAN bits, Front Wipers On and Front Wiper in Intermittent position, are set by BEM on message \$403. The Front Wiper in Intermittent position, Bit 55, is set to '1' when the BEM detects the front wiper switch is in the intermittent position. Front Wiper Operating, Bit 19, is set to '1' when the BEM detects that front wipers are operating due to switch being in Intermittent or positions I or II. This bit also set when the BEM is performing a wipe after wash function.

4.4.4 Diagnostics

The following State Encoded Diagnostic PIDS are used:

- PID \$1912 – Transmission Gear Selected (PRNDL) – SAE (State: Reverse)

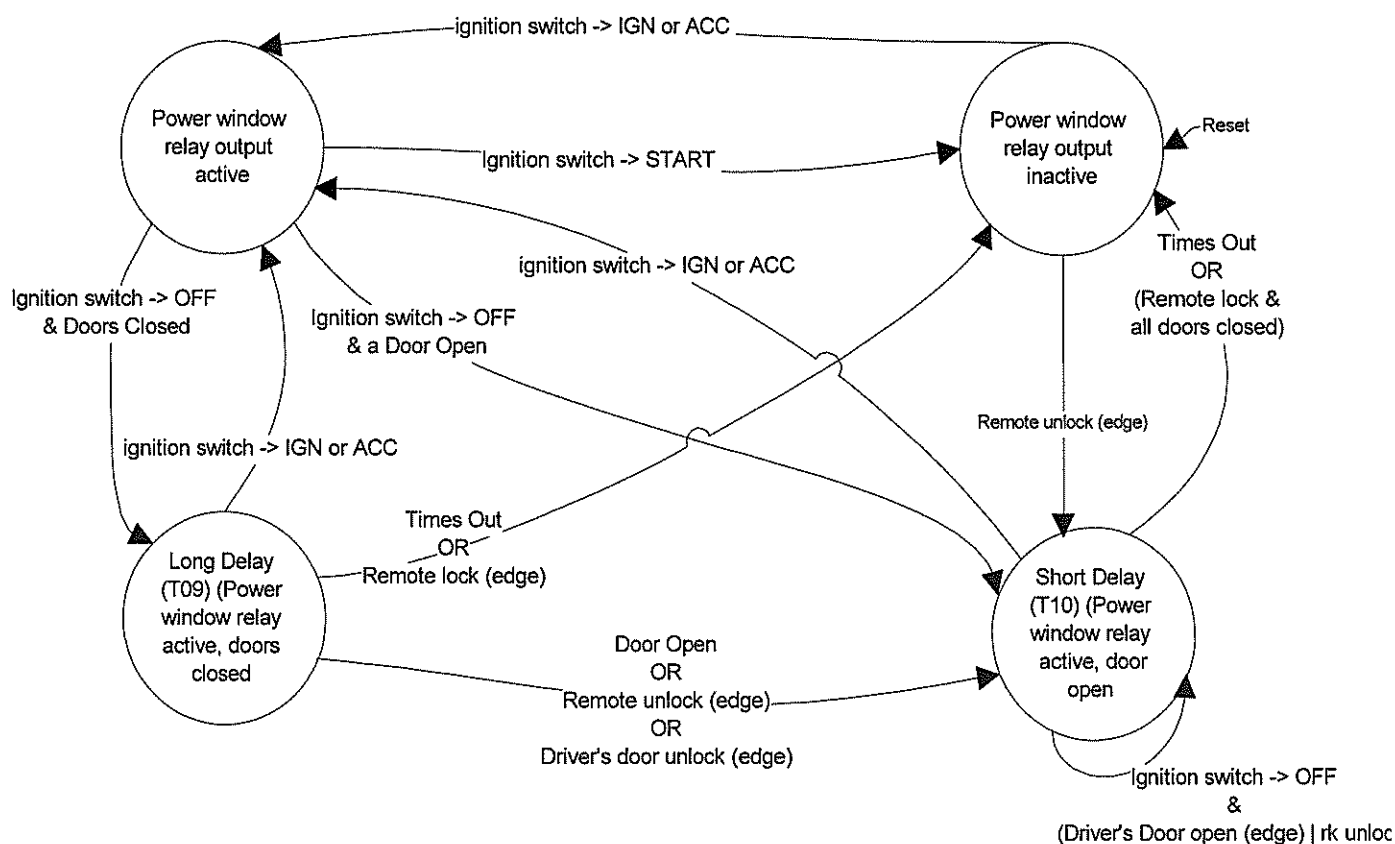
The following Bit Mapped Diagnostic PIDs are used:

- PID \$A448 – Rear Wiper Input Switch Status (Bits: Rear Wiper Input Switch Status, Rear Wiper Washer switch Status)
- PID \$A459 – Rear Wiper/Washer Output Status (Bits: Rear Wiper Output Status)
- PID \$C137 – Liftgate & Liftgate Glass Status (Bits: Liftgate Glass State, Liftgate Glass Switch)

4.5 Power Window Control

With the ignition switch in the accessories or ignition position, the power window relay output is activated (VOL). Switching the ignition switch to the start position will deactivate the power window relay (VOH). Switching the Ignition switch to the off position will allow the power window relay output to remain active for T09 before deactivating. However if during this time, a door is opened, then the time is reduced to a shorter period (T10), starting from when the door was opened, to ensure compliance with Australian Design Rule 42. Power window relay output is also activated for remote unlock and stays active for T10 before deactivating. If during this time (T10) the drivers door is opened then the power window relay output remains active for another T10 (from when door is opened) before deactivating.

Power window relay output is also deactivated for all doors closed and remote lock requested.



State diagram for power window control

For any discrepancies, the functional description supersedes the above state diagram

From	Activation Method	To
Reset		Inactive Mode
Inactive Mode	Ignition switch-> IGN or ACC	Active Mode
Inactive Mode	Remote unlock (edge)	Short exit delay Mode
Active Mode	Ignition switch -> START	Inactive Mode
Active Mode	(Ignition switch -> OFF) & (doors closed)	Long exit delay Mode
Active Mode	(Ignition switch -> OFF) & (a door open)	Short exit delay Mode

From	Activation Method	To
Long exit delay Mode	Ignition switch-> IGN or ACC	Active Mode
Long exit delay Mode	Times Out (T09)	Inactive Mode
Long exit delay Mode	Remote lock (edge)	Inactive Mode
Long exit delay Mode	A Door open	Short exit delay Mode
Long exit delay Mode	Remote unlock (edge)	Short exit delay Mode
Long exit delay Mode	Driver's door unlock (edge)	Short exit delay Mode
Short exit delay Mode	Ignition switch-> (IGN or ACC)	Active Mode
Short exit delay Mode	Times Out (T10)	Inactive Mode
Short exit delay Mode	Remote lock & all doors closed	Inactive Mode
Short exit delay Mode	(Ignition switch -> OFF) & (Driver's door open(edge))	Short exit delay Mode
Short exit delay Mode	(Ignition switch -> OFF) & (remote unlock (edge))	Short exit delay Mode

4.6 Sunload Sensor Module Interface

Activation of the battery saver supply also provides power to the sunload/twilight sensor module. With the ignition switch in the ignition position and the battery saver supply active, the BEM reads the single sunload sensor signal and transmits the raw data representing driver and passenger sunload readings over the CAN bus for benefit of the climate control system. Invalid data is signalled with the Ignition switch in off or accessories on position or when a valid sunload reading is not available due to a fault. With Ignition on a fault condition will always default headlamp relays on if in Auto position.

The Input voltage, VSUNLOAD, is the voltage seen at the connector pin of the BEM.

The formula to calculate the A/D value depending on the regulator voltage is:

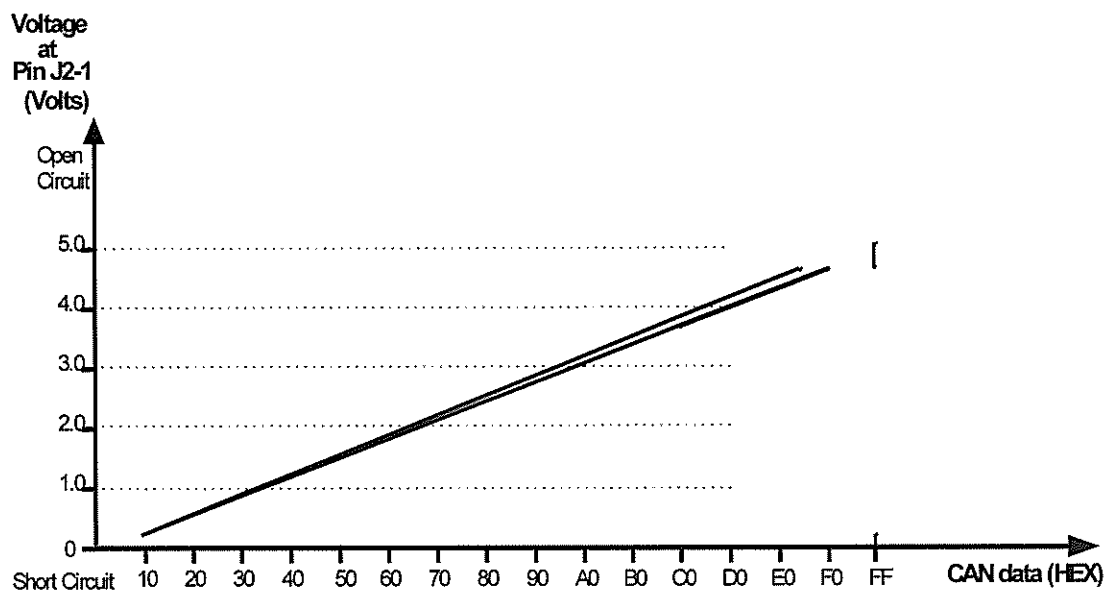
$$A/D = (256 \times VSUNLOAD / VREGULATOR) - 1 \quad \text{for } (VSUNLOAD > 0.2V \text{ and } VSUNLOAD < 4.7V)$$

$$VSUNLOAD = \text{Invalid data, } TCAN = \$FF \quad \text{for } (4.9 \leq VREGULATOR \leq 5.1)$$

Note: Sunload Sensor Tolerances is $\pm 2.5\%$ (does not consider additional $\pm 2.5\%$ due to Sun Sensor Module)

VSUNLOAD (Volts)	A/D (Hex)	TCAN (Hex)	TCAN Dec)	VSUNLOAD (Volts)	A/D (Hex)	TCAN (Hex)	TCAN (Dec)
5.1	FF	FF	255	2.5	7C-81	7C-81	124-129
5.0	F9-FF	FF	255	2.4	77-7C	77-7C	119-124
4.9	F4-FF	FF	255	2.3	72-77	72-77	114-119
4.8	F4-F9	FF	255	2.2	6D-71	6D-71	109-113
4.7	F4	FF	255	2.1	68-6C	68-6C	104-108
4.6	E5-F3	E5-F3	229-243	2.0	63-67	63-67	99-103
4.5	E0-EA	E0-EA	224-234	1.9	5E-62	5E-62	94-98
4.4	DB-E4	DB-E4	221-228	1.8	59-5D	59-5D	89-93
4.3	D6-DF	D6-DF	214-223	1.7	54-57	54-57	84-87
4.2	D1-DA	D1-DA	209-218	1.6	4F-52	4F-52	79-82
4.1	CC-D5	CC-D5	204-213	1.5	4A-4D	4A-4D	74-77
4.0	C7-CF	C7-CF	199-207	1.4	45-48	45-48	69-72

VSUNLOAD (Volts)	A/D (Hex)	TCAN (Hex)	TCAN (Dec)	VSUNLOAD (Volts)	A/D (Hex)	TCAN (Hex)	TCAN (Dec)
3.9	C2-CA	C2-CA	194-202	1.3	40-42	40-42	64-66
3.8	BD-C5	BD-C5	189-197	1.2	3B-3D	3B-3D	59-61
3.7	B8-C0	B8-C0	184-192	1.1	36-38	36-38	54-56
3.6	B3-BB	B3-BB	179-187	1.0	31-33	31-33	49-51
3.5	AE-B5	AE-B5	174-181	0.9	2C-2E	2C-2E	44-46
3.4	A9-B0	A9-B0	169-176	0.8	27-28	27-28	39-40
3.3	A4-AB	A4-AB	164-171	0.7	22-23	22-23	34-35
3.2	9F-A6	9F-A6	159-166	0.6	1D-1E	1D-1E	29-30
3.1	9A-A0	9A-A0	154-160	0.5	18-19	18-19	24-25
3.0	95-9B	95-9B	149-155	0.4	13	13	19
2.9	90-96	90-96	144-150	0.3	0E-0A	0E-0A	14-10
2.8	8B-91	8B-91	139-145	0.2	09	FF	255
2.7	86-8C	86-8C	134-140	0.1	04	FF	255
2.6	81-86	81-86	129-134	0.0	00	FF	255



4.7 Courtesy Lamp, and Batter Saver Supply Control

4.7.1 Courtesy Lamp Operation with Illuminated Entry and Exit Delays

While the ignition switch is in the accessories, ignition or start position, and any door (including boot for wagon/Territory) ajar status indicates that a door is open, then the courtesy lamp output will activate (VOL) with a short linear fade-in (T13). With the door (including boot for Territory/Wagon) closed, it will deactivate (VOH) with a short linear fade-out (T13).

While the ignition switch is in the off position, and while a door (including boot for Territory/Wagon) remains open, the courtesy lamp will remain active until the battery saver timeout has expired (T11 in Normal Mode and T12 in plant mode or intermediate plant mode). When all open doors (including boot for Territory/Wagon) are closed, an exit delay shall occur, during which the courtesy lamp will remain active for a short time of T14. It will then fade linearly to the deactivated state over a period of T15. This exit delay shall be overridden by an open door (including boot for Territory/Wagon), turning the ignition out of the off position, or by a remote key activation, or pressing of the dome lamp switch.

If the ignition key is removed from the ignition switch (i.e.: removed from the transponder magnetic field) within a time (T11 in Normal Mode, T12 in Plant Mode) of turning the ignition switch to the off position and with no door (including boot for Territory/Wagon), having been opened, then an exit delay shall occur. This function may be disabled at RBAU plant. If disabled this will override transponder related dome lamp functions.

While the ignition switch is in the off position, activation of the panic alarm by momentarily pressing the remote key 'Panic' button shall activate the courtesy lamps. Activation of the remote 'Door Unlock', 'Boot Release', or deactivation of the panic function by either the remote key or by time out shall turn on courtesy lamps causing an entry delay to occur, during which time the courtesy lamp will remain active for a short time of T14. It will then fade linearly to the deactivated state over a period of T15. This delay shall be overridden by an open door (including boot for Territory/Wagon), turning the ignition out of the off position, or by a remote key activation, or pressing of the dome lamp switch.

While the ignition switch is in the off position, momentarily pressing the remote key 'Door Lock' button to lock the doors, while they are closed will deactivate (VOH) with a short linear fade-out (T13).

If no remote key has been taught (i.e.: in plant mode or intermediate plant mode); then the courtesy lamp will deactivate after a shorter time than usual, T12, when the doors (including boot for Territory/Wagon) are open, regardless of the position of the ignition switch.

When the courtesy lamp turns on, it will have a short linear fade-in (T13). Likewise, when lamp turns off, it will have a linear fade-out (T13). If any fade-in or fade-out period is interrupted by a change of state, then the fade shall start from the current brightness level and only last the time it takes to complete the action.

The courtesy lamp switch via CAN can be used to turn on and off the courtesy lamp, independent of other conditions. BEM will act on positive edge of the courtesy lamp switch.

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4.7.2 Battery Saver Supply

When the battery saver supply is active (VOES), it allows the courtesy lamps, the boot lamp and the glove box lamp to illuminate. Also the TILA supply shall also be active to ensure keyhole illumination. Battery saver also provides power to sun load sensor and antenna for audio.

The battery saver supply is activated when the ignition switch is in the accessories, ignition or start position. The battery saver supply is also activated by the opening or closing a door/boot, or pressing the remote key 'Door Unlock', 'Door Lock', 'Boot' or 'Panic' button.

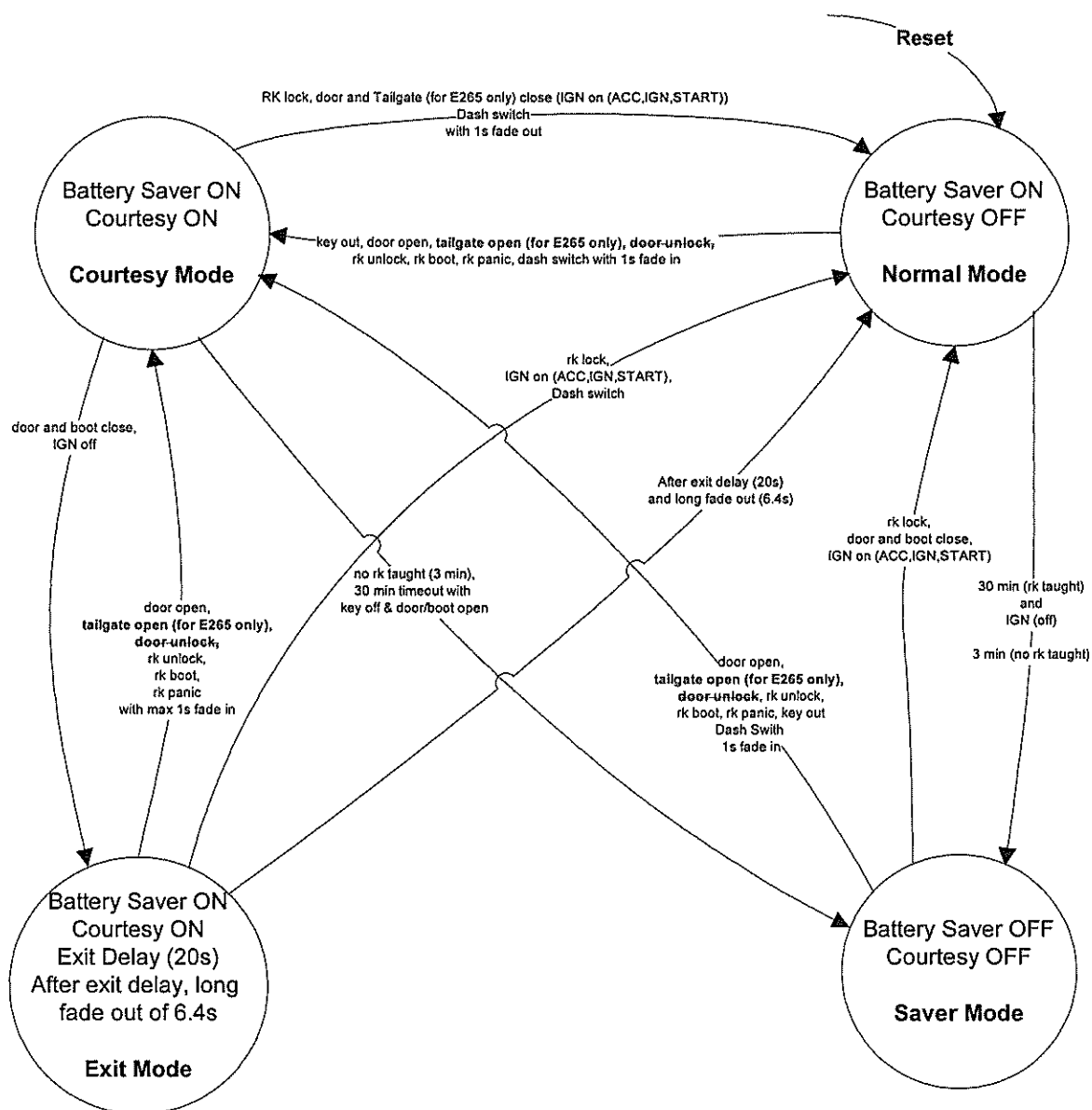
The battery saver supply is deactivated (VOUS) after a time (T11) with the ignition switch being turned to the off position as long as the last event of opening or closing a door/boot or pressing any remote key button has occurred.

If no remote key taught (i.e.: in plant mode or intermediate plant mode), the battery saver supply will deactivate after a shorter time period of T12 regardless of the position of the ignition switch. The battery saver timeout is re-enabled whenever the ignition switch is turned between the off and accessories positions, a door/boot is opened or closed, or a remote key door or boot release functions are operated.

If the Battery saver supply is deactivated, and Audio Keyless Play is activated, then if BEM is not in remote key plant mode, the Battery Saver can be re-enabled by bit 24 in CAN message 307 (illumBatsaverRequest) from audio module.

The table below shows six states for the courtesy lamp and battery saver supply.

State	Summary of State
Normal Mode	Battery Saver ON, Courtesy OFF
Saver Mode	Battery Saver OFF, Courtesy OFF
Courtesy Mode	Battery Saver ON, Courtesy ON
Exit Mode	Battery Saver ON, Courtesy ON. This consists of T14 exit delay and T15 fade out.
Audio Keyless Play	Battery Saver ON, Courtesy OFF or ON
Police Mode	Depends on the configuration

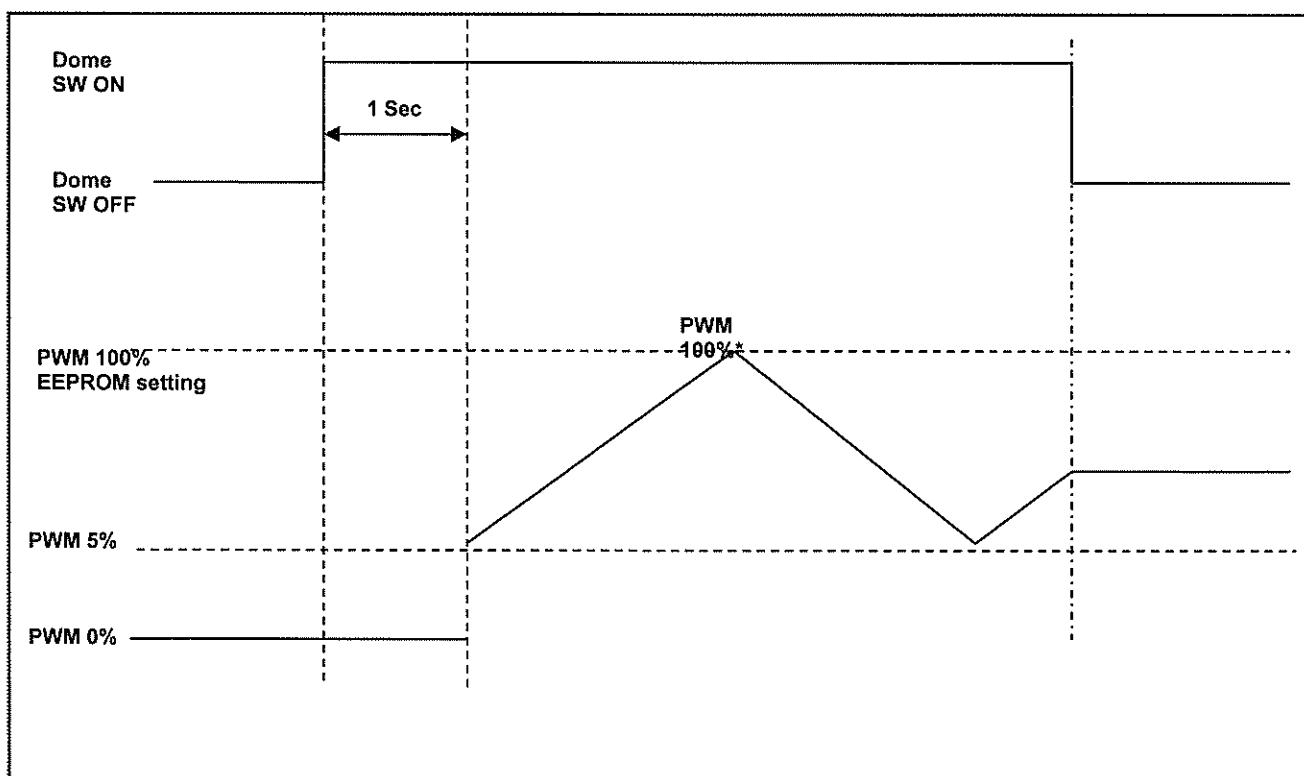


State diagram for courtesy lamp and battery saver supply control
 For any discrepancies, the functional description supersedes the above state diagram

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4.7.3 Mood Lighting

To enter mood lighting control, dome lamp dash button needs to be pressed for T79. After it the dome will turn on at pwm value of 5% and increase at rate of 1% per T83 of switch press to a maximum of pwm value of 100%. Then it will start to turn off at the same rate. The figure below shows an example of dome lamp pwm ramp up and down due to dome lamp switch press. Releasing the control after a press and hold causes the dome lamp to stay on at that intensity. The new intensity setting is retained in memory and is recalled each time there is a short press of the dome lamp control or the doors are opened enabling the lighting. The default value for dome lamp intensity is 100%.



A short press of the dome lamp switch (<2 seconds) turn the lamp On or Off (if already on).

4.8 Central Door Locking Related Functions

4.8.1 Door and Boot Ajar Status

The BEM reports the door ajar status for all four individual doors at all times, subject to the conditions for CAN message transmission. The BEM also reports the rear tailgate/rear window ajar signal.

4.8.2 Manual (and Remote) Tailgate and Rear Window Release

Mechanical opening (with mechanical key) of rear tailgate will not affect lock status of other doors.

Rear tailgate and rear window (for Territory) buttons are enabled during the following conditions:

- When driver's door is unlocked (Plant Mode).
- When passenger's door is unlocked (Normal Mode)
- Remote button for tailgate pressed.

Rear tailgate and rear window (for Territory) buttons are disabled during the following conditions:

- If tailgate and rear window(for Territory) buttons are enabled due to remote button press:
 - When timeout of T69 expires and passenger doors are in locked state.
 - When rear tailgate/window ajar and time T75 passed since last tailgate or rear window switch is activated and passenger doors are in locked state.
- When vehicle speed is greater than T70.
- When vehicle is locked (via mechanical or remote)
- *When the Battery over Voltage Level is detected.*

Read body configuration using ReadDataByCommonIdentifier service (\$22) with PID equal to \$E11A.

State	Description	Comment
00	Unknown Body Type	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock (and power will be supplied to tailgate and window (for territory)) and horn alert will sound. If the 5th door input is active when a remote lock is requested, then the horn warning will sound and other doors will remain locked.
01	Sedan	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock (and power will be supplied to tailgate and window (for territory)) and horn alert will sound. If the 5th door input is active when a remote lock is requested, then the horn warning will sound and other doors will remain locked.
02	Wagon	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock (and power will be supplied to tailgate and window (for territory)) and horn alert will sound. If the 5th door input is active when a remote lock is requested, then the horn warning will sound and other doors will remain locked.
03	Utility	
04	LWB	
05	Territory	If a door (excluding 5th door) is ajar and remote lock is requested, all doors will unlock (and power will be supplied to tailgate and window (for territory)) and horn alert will sound. If the 5th door input is active when a remote lock is requested, then the horn warning will sound and other doors will remain locked.
06	Coupe	
07	Van	

Body type is configured using WriteMemoryBlock service (\$3B) for Block \$00. RBAU plant will set the body type as \$00 to identify the module as not been configured.

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4.8.3 Secure Tailgate Strategy (For Orion Only)

When Secure Boot is enabled and vehicle exited and locked after the ignition has been turned off, the following will occur:

- Tailgate/Window release is disabled if driver door remain locked.
- Tailgate/Window release is disabled if driver door unlocked or ajar.
- Tailgate/Window release is enabled if ignition is on, vehicle is mobilised, and the vehicle speed is below T70.

In DST plant mode, secure boot is disabled.

4.8.3.1 Enable/Disable Secure Boot

The ICC "Secure Boot" feature must be selected in graphical interface for the ICC. This enables a CAN message from ICC screen to BEM indicating the secure boot feature is active. Secure Boot setting can be changed (enabled/disabled) only when ignition is on and vehicle is mobilised.

4.8.4 Manual Central Door Locking Functions

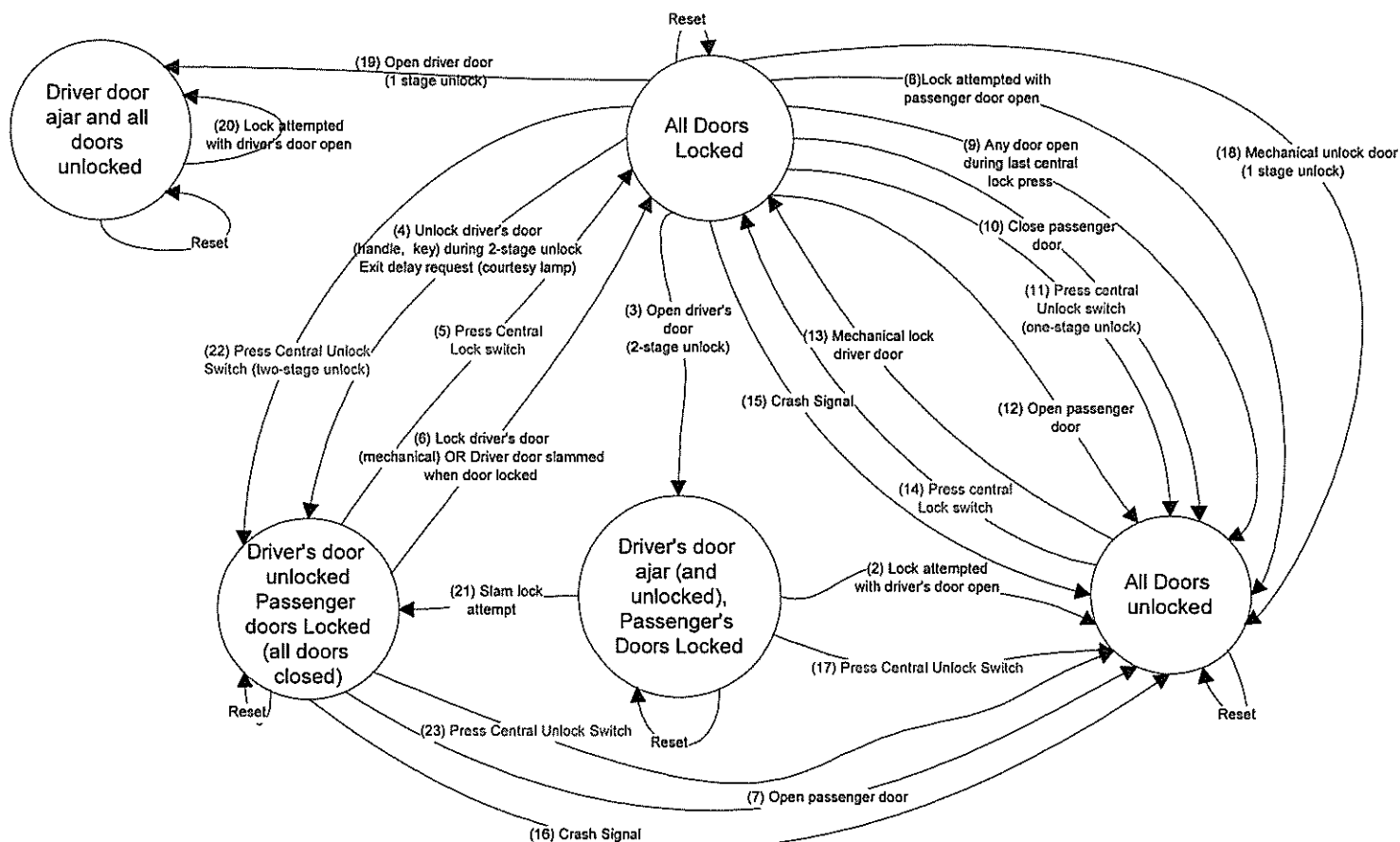
Manual central locking operation may be initiated by pressing the central door locking switches (lock and unlock) via CAN (momentary); manually operating the driver's door lock mechanism by using a DST key, soldier button or interior door handle; or by opening a door.

A crash (identified by the air bag deployed signal via CAN) triggers an all door unlock action. Once a crash unlock has occurred, then subsequent door switch inputs shall be ignored including the manual Central Locking switch bit via CAN, until the next ignition switch cycle occurs.

When a slam door is detected, unlock driver's door only, unless a passenger door is open. The driver's door is detected as slammed, if it has been just closed and the door is locked.

When driver's door is ajar and central lock switch is pressed or soldier button of driver's door is pressed, all door unlock will be activated even if passenger doors are in unlocked state.

After power up or a reset, lock status will not be changed to avoid lockout risks or doors unlocking on power up, and it is assumed that passenger doors are in locked state.



State diagram for door lock operation

For any discrepancies, the functional description supersedes the above state diagram

4.8.5 Mechanical Door Lock Strategy

During single stage unlock mode, based on last remote key pressed:

- Open driver door from inside using the interior handle will unlock all doors.
- Mechanically unlocking driver door (with mechanical key) will unlock all doors.

During two-stage unlock mode, based on last remote key pressed:

- Opening the driver door mechanically (using interior handle or a mechanical key from outside) will have no effect on passenger doors.

Operation of the Central Locking switch does not change independent of the BEM being in one stage or two stage unlock modes.

Operation is summarised in the following table (Refer Sect 0 for Taxi Mode operation):

Action	Activation Methods (Transitions)	Output		Time at VOUS	Time at VOES
Driver's door unlocks	• Mechanically unlock driver's door, with all doors previously locked, during 2 stage unlock (4)*. • Open driver's door, during 2 stage unlock when all doors are locked (3)*. • Lock attempted with driver's door open (2). • Driver's door slammed when door locked (21). • Press central unlock switch with two-stage unlock mode(22).	Driver's door unlock line		—	T19
All doors unlock	• Press central door unlocking switch with driver's door opened and passenger door locked (17). • Mechanically unlock driver's door, with all doors previously locked, during 1 stage unlock (18) • Press central door unlock switch with driver's door locked and one-stage unlock mode (11). • Open a passenger's door with driver's door locked (7) or (12)*. • Lock attempted (driver door) with a passenger door open (8). • Passenger door is closed with driver's door locked (10). • Crash indicated (Air Bag deployed via CAN) (15) or (16). • Press central door unlock switch with driver's door unlocked (23).	Door unlock line		—	T19
		Driver's door unlock line **		—	T19
Driver's door ajar and all doors unlocked	• Open driver door, during one stage unlock, with a mechanical key (19). • Lock attempt with driver's door open (20)	Door unlock line		—	T19
		Driver's door unlock line **		—	T19
All doors lock	• Press central door locking switch with driver's door unlocked and closed (5) or (14). • Mechanically lock driver's door (6) or (13)	Door lock line		—	T19

** If driver's door is already unlocked, then driver's door unlock line is not activated.

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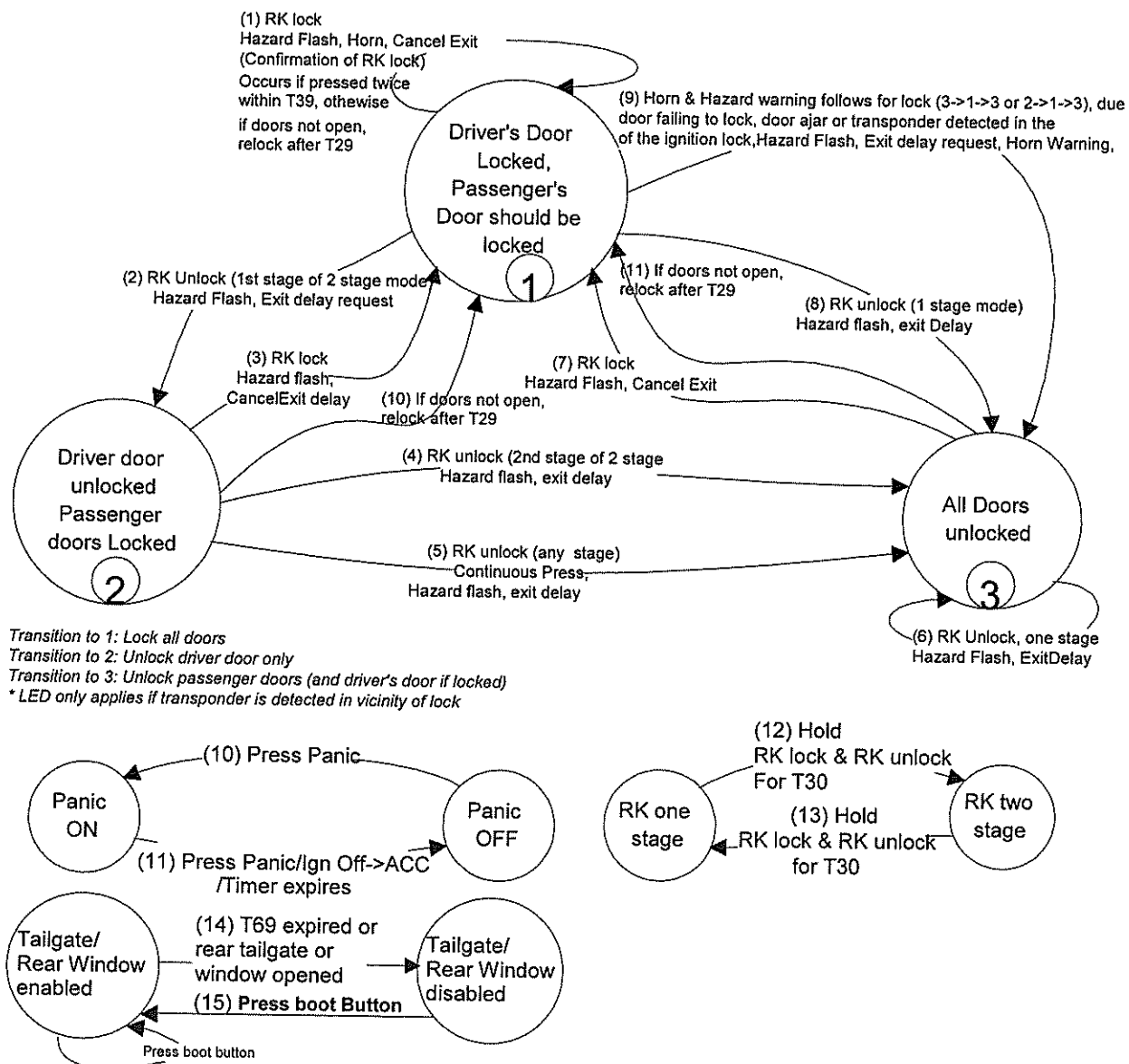
4.8.6 Remote Central Locking Functions with Rear tailgate/window enable

Remote entry and Panic alarm functions operate when the ignition switch is in the off position, although the RF receiver is active at all times. A routine to distribute EEPROM wear is employed for frequently changing bytes (rolling code counters).

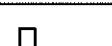
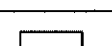
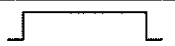
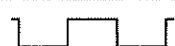
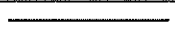
Two-stage unlocking may be activated or deactivated for each remote key. If one-stage unlocking is active, then a single press of the 'Door unlock' button shall unlock all doors. If two-stage unlocking is active, then only the driver's door shall unlock on the first remote key press of the 'Door unlock' button. The remaining doors shall open in a second press, or if the button is still held down once the first press is complete. To toggle between one stage and two stage unlocking, press both remote key "door lock" and "door unlock" buttons simultaneously for T30. The status for each remote key is stored in EEPROM and defaults to one-stage mode. Changing modes is the only instance where multiple button presses are acknowledged.

Pressing the 'Panic' button activates the panic alarm. If pressed, the panic alarm shall operate until the ignition is switched out of the off position; the button is pressed a second time, or the panic alarm period times out (T26). Hazard lamp flashing is activated via CAN. The software includes measures to reduce wear on the driver's door lock motor by not unlocking it when it is already unlocked. Hazard lamp flashing is activated via CAN.

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State diagram for remote central locking functions
 For any discrepancies, the functional description supersedes the above state diagram

Action	Activation Methods (<i>Transitions</i>)	Output		Time at Vous	Time at VoES
Remote driver door unlock	Press the remote 'Door Unlock' button when driver's door is locked with two stage unlocking active (2).	Driver's door unlock line		—	T19
		Hazard lamps via CAN		—	T74
Remote all doors unlock	Press the remote 'Door Unlock' button when driver's door is locked with one stage unlocking active (8).	Driver's door unlock line		—	T19
		Other door unlock line		—	T19
		Hazard lamps via CAN		—	T74
Remote passenger doors unlock	- Press the remote 'Door Unlock' button with the driver's door unlocked and two stage unlocking active (4), (6). - Continue to hold the remote 'Door Unlock' button after the driver door has unlocked for 2 seconds with two stage unlocking active (5), (6).	Other door unlock line		—	T19
		Hazard lamps via CAN		—	T74
Remote door lock	Press the remote 'Door Lock' button when the driver's door is unlocked (3), (7).	Door lock line		—	T19
		Hazard lamps via CAN		T72	T71, T73
Confirmation of remote door lock	Press the remote 'Door Lock' button, when all doors are locked. Horn occurs only on second press within T39. (1).	Door lock line		—	T19
		Hazard lamps via CAN		T72	T71, T73
		Horn relay		—	T24
Enable Tailgate / Tail Window release switches.	Press remote 'Boot' button (15).	Tailgate / Window switch enabled.		—	T69
		Hazard lamps via CAN		—	T21
Panic alarm activation	Press the remote 'Panic' button (10).	Hazard lamps via CAN		T27	T27
		Horn relay		T27	T27
Panic alarm deactivation	- Press the remote 'Panic' button (11). - Turn the DST ignition key out of off position (11). - Panic alarm timer expires (11).	Hazard lamps via CAN		—	—
		Horn relay		—	—

Toggle between one stage and two stage unlocking	- Press both 'Door Lock' and 'Door Unlock' buttons simultaneously for T30 (12), (13)	Hazard lamps via CAN		T20	T71, T73
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Note: Multiple remote key presses are ignored except for mode changes.

4.8.7 Remote Lock Horn and Hazard Warning

If the remote key is used to lock the vehicle, a number of checks are made to ensure it is secured. A horn warning shall sound if the BEM detects an open door, tailgate or tail window, an unlocked driver's door, or if a transponder is in the vicinity of the steering lock. This last cause would mean that either a DST ignition key had been left in the lock, or that a transponder had been left near the lock for the purposes of theft while the vehicle is unattended. If the warning is due to a transponder detected by the steering lock, then the CAN message to activate the LED, will also remain on. The CAN message, activating the LED, reverts back to normal operation if a subsequent remote lock is successful, or if the ignition switch is turned out of the off position.

Activation Method	Output		Time at Vows	Time at VOES
Remote door lock where a door is ajar (open), a door fails to lock or a transponder is detected in the vicinity of the ignition switch <i>(Transition 9)</i>	Door lock line		—	T19
	Door unlock lines (a door ajar or door fails to lock)		—	T19
	Door unlock line(driver) (transponder detected)			T19
	Hazard lamps via CAN		T72, T78, T23	T71, T73, T23, T74
	Horn Warning relay		T25	T24
	CAN message to activate LED output (if BEM detects a transponder)		—	—
	Courtesy lamp output (if BEM detects a transponder)		—	Exit delay

4.8.8 Automatic Door Re-lock

If the remote key is used to unlock the vehicle (momentary) and then no door is opened including tailgate (or tail window for Territory), the driver's door is not locked, or the ignition switch is not turned, then the doors will automatically re-lock after a period of time, T29.

4.8.9 Drive-Away Locking

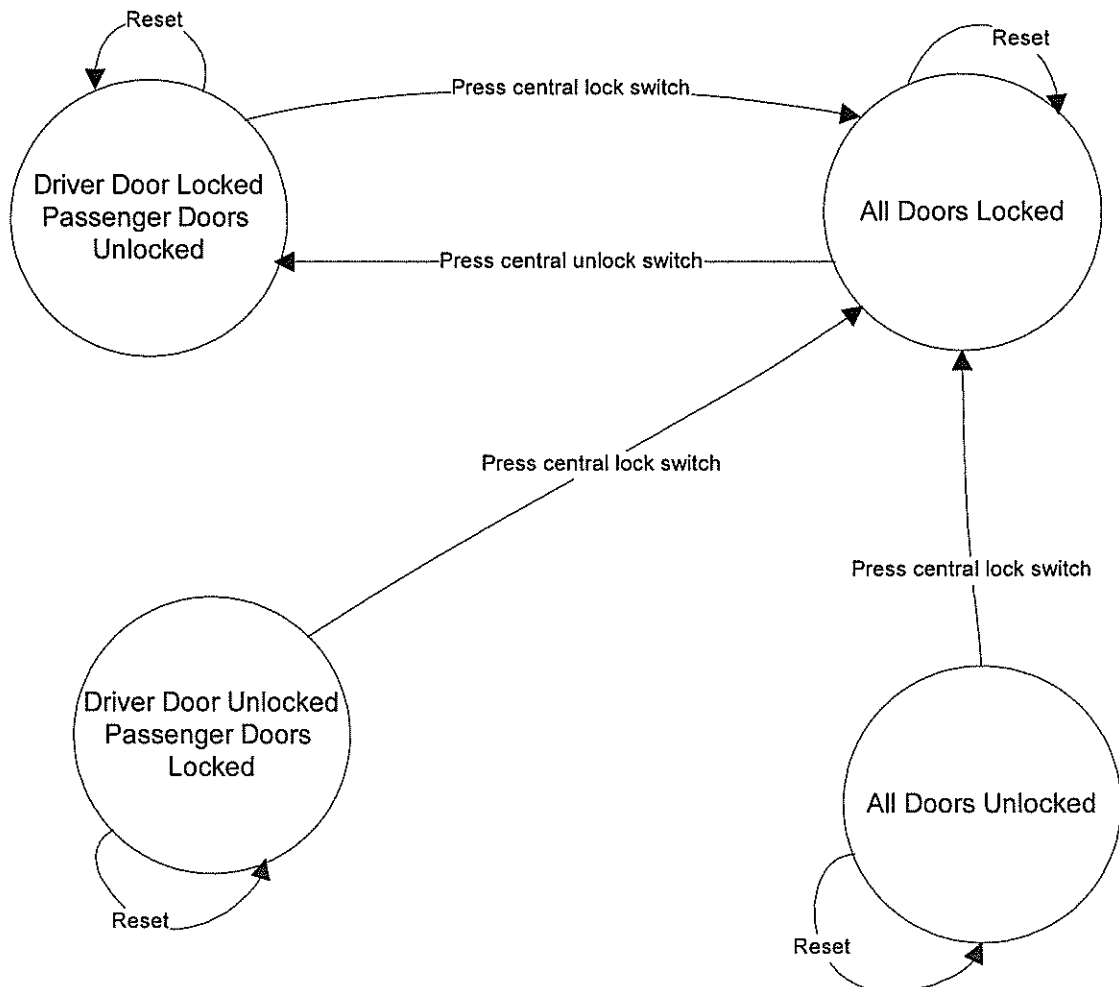
When ignition switch in the RUN position, all doors will lock when the vehicle speed increases above T70 and driver or passenger door is unlocked.

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4.8.10 Taxi Mode

One press of central door lock dash switch will lock all doors. One press of central door unlock dash switch will unlock only the passenger doors and enable tailgate and rear window release switches. Refer state diagram for door lock status in Taxi Mode for more details.

After system reset, the BEM software checks the passenger door ajar inputs (this does not include rear tailgate/window ajar input). If any passenger door is ajar, the software will assume the passenger doors are unlocked.



State diagram for door lock status in Taxi Mode

For any discrepancies, the functional description supersedes the above state diagram

4.9 RF Remote Control Related Functions

The main function of this module is to recognise a control message from a trained remote key, and pass on the recognised key values and key edges to other software modules. In addition, it also provides key training function for matching with the right remote key and resynchronisation criteria if the trained remote key falls out of the synchronized range.

4.9.1 RF Control Features

RF signals are received by the BEM, and responds to remote keys that conform to the Shark Protocol (Refer 3.3.4). In addition to detecting button presses for remote entry functions, remote key training and resynchronisation features are included.

BEM will only respond to a trained remote key when ignition is OFF, or during remote key training process when the ignition switch is in the accessory position.

In plant mode, the BEM responds only to a master remote key (TIC code \$A0 00 00 00 with rolling code disregarded). Additional remote keys cannot be taught to the BEM whilst in plant mode. After exiting from plant mode normal remote key teaching can be performed. Once the master remote key is erased from memory, it can never be re-taught to a BEM. Master remote keys are only produced for the use of Bosch and Ford plants.

The Barra BEM only responds to 5 different control messages from a trained 433MHz RF remote key, which sends out SHARK protocol. They are

- Door Lock
- Door unlock
- Tailgate/Boot release
- Panic
- (Door lock) + (Door unlock)

BEM will only respond to trained key when ignition is OFF, or during key training process when ignition is in the accessories position.

4.9.2 RF Controlling Codes

All control codes only operate when the ignition switch is at the off position.

Lock all doors.

- Door lock (control code = 0x10)

Unlock all doors if one stage unlocking is activated. Only unlock driver's door from all door locked if two stage unlocking is activated, the next press to the unlock key (or holding unlock key for 2 seconds) will unlock all doors.

- Door Unlock (control code = 0x20)

Release tailgate/boot.

- Tailgate/boot release (control code = 0x40)

Initiate/stop car alarm system.

- Panic (control code = 0x08)

Toggle between 1 stage door unlocking system and 2 stages door unlocking system.

- Door Lock key together with Door unlock key (control code = 0x30)

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Table A

Key Icon	Control code in hexadecimal form	Control code in binary form
Lock	0x10	0001 0000
Unlock	0x20	0010 0000
Tailgate/Boot Release	0x40	0100 0000
Panic	0x08	0000 1000
Lock + Unlock	0x30	0011 0000

4.9.3 Counter Bytes

The counter is two bytes of codes that will be incremented with each separate button activation. The bits of these two bytes will be interleaved to form the rolling code bytes used for encryption. The counter bytes provide the functions of

- ensuring 1 action per button pressed (if required)
- encrypting the RF message

With continuously updating to this counter, it is the best option to use this counter as a random number source for message encryption. Refer to document RKE Protocol SDS – Release 1.3 for description of data encryption.

The counter bytes are written to the EEPROM of the BEM. To avoid wearing out of the EEPROM, the counter byte will be written to EEPROM once in 32 remote key presses. Within these 32 presses, the counter bytes are written to the volatile RAM of the microprocessor.

4.9.4 Remote Key Teaching

In other cases (plant mode, intermediate plant mode and normal operating mode), switching ignition switch from the off position to the accessories position activates a short time window (T32). If three momentary activations of the rear demist switch are detected in this time, Remote Key Teach Mode is entered. Acknowledgment that the system has entered Teach Mode is via the mode acknowledgment sequence (see section 4.14.2 for further details). Dealer may use a diagnostics command that is available to simulate the pressing of the switch. To use this command, the ignition switch must be placed in the accessories position before the command is sent.

Once teach mode has been entered, all remote key codes stored in the BEM's non-volatile memory will be cleared. Upon receipt of a valid remote key code, the system will write this code to non-volatile memory and acknowledge receipt of this code. Receipt of an already acknowledged code will result in no action. A maximum of eight unique remote keys may be taught. Teach mode will be exited if the ignition switch is turned out of accessories into the off or ignition position. Acknowledgement that training mode has been exited is via the mode acknowledgement sequence.

4.9.5 Plant Mode

In Plant Mode, the BEM responds only to a Master Remote Key (code \$A0 00 00 00 as TIC), additional Remote Keys cannot be taught to the BEM whilst in Plant Mode.

Master remote key only operates when the BEM is in plant mode. After exited from plant mode, the master remote key cannot operate any function of the BEM, nor can the Master remote key be trained as an operating remote key. The TIC of the Master remote key (\$A0 00 00 00) is considered to be unique for Master remote key and will not be used for any other remote keys. Master remote key does not need to be re-synchronised. BEM will respond with an action, to a signal from a Master remote key, with a different rolling code.

4.9.6 Receiver Resynchronization

The remote entry receiver is capable of passively resynchronizing to a remote key in one of two ways if they become out of synchronization. The remote key is considered out of synchronization if it is more than 1024 counts greater than the receiver. A receiver may passively resynchronize to a single programmed remote key if the remote-key's counter is less than 45,055 counts greater than that of the receiver, the TIC matches and three transmissions are received in a row, with the transmitted counter incrementing only one time with each transmission. A receiver may also passively resynchronize if a recognized out of synchronization remote key sends a transmission within 30 seconds of a different valid remote key's transmission.

4.9.7 Intermediate Plant Mode

In the first stage plant mode, a Master Remote Key will be able to control all vehicles installed with a BEM set to plant mode. To prevent Master Remote Key access to a plant-mode BEM installed vehicle, an intermediate plant-mode is set up. In this mode, access to the car has to be done through the mechanical key. The BEM will enter intermediate plant-mode state by setting of a diagnostic PID via the CAN line. Once this intermediate plant-mode state is entered, it can never be reversed back to the first state plant mode.

4.9.7.1 Entering Intermediate Plant Mode

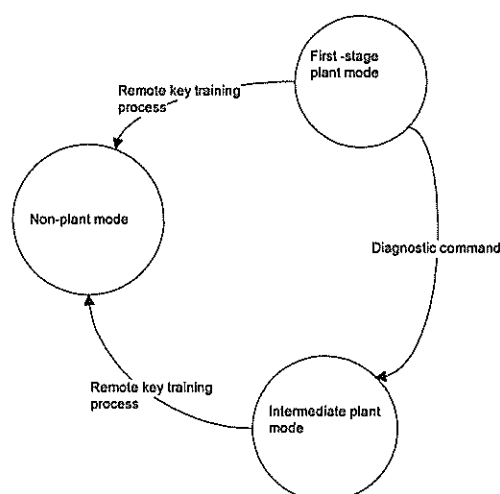
When BEM receives diagnostic command WriteDataByCommonIdentifier (\$2E) with PID \$C901 set to \$01 via the CAN bus, the BEM will then enter this intermediate plant mode. PID \$C901 cannot be set to any other values by WDS. PID \$C901 will be ignored during normal operating mode.

4.9.7.2 Differences between first stage Plant Mode and Intermediate Plant Mode

In the intermediate plant mode, BEM is no longer under Master Remote Key control; all other plant mode features remain unchanged.

4.9.7.3 Exiting Intermediate Plant Mode

To exit the intermediate plant mode, normal remote keys (not the Master Remote key) training mode has to be involved. Remote key can be trained to the BEM using the standard training procedure or through diagnostic control. Once the teaching process is completed, the BEM is considered to have come out of the intermediate plant mode and entered normal operating mode. The same key teaching process will also get the BEM to come out of plant mode from first stage plant mode, thus skipping the intermediate plant mode process. After entering key training process, BEM will not get back to Intermediate plant mode with the use of WDS tools alone. Training zero remote key to the BEM will bring the BEM to intermediate plant mode.



Intermediate Plant Mode State diagram

For any discrepancies, the functional description supersedes the above state diagram

4.10 Door Lock Status

The following messages are updated due to a door lock or unlock:

	Bit	Description
Msg \$403 BEM_Status_MS	RFD_Locked	'1' -Right Front Door Locked '0' -Right Front Door Unlocked
	LFD_Locked	'1' -Left Front Door Locked '0' -Left Front Door Unlocked
	RRD_Locked	'1' -Right Rear Door Locked '0' -Right Rear Door Unlocked
	LRD_Locked	'1' -Left Rear Door Locked '0' -Left Rear Door Unlocked
	Key_1_LockRequest	'1' - Remote key 1 lock request. '0' - Remote key 2-8 lock press or any remote key unlock press.
	Key_2_LockRequest	'1' - Remote key 2 lock request. '0' - Remote key 1 or 3-8 lock press or any remote key unlock press.
	Key_1_UnlockRequest	'1' - Remote key 1 unlock request. '0' - Remote key 2-8 unlock press or any remote key lock press.
	Key_2_UnlockRequest	'1' - Remote key 2 unlock request. '0' - Remote key 1 or 3-8 unlock press or any remote key lock press.

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4.11 Memory Seat and Priority Key Control

4.11.1 Signalling in Normal Mode

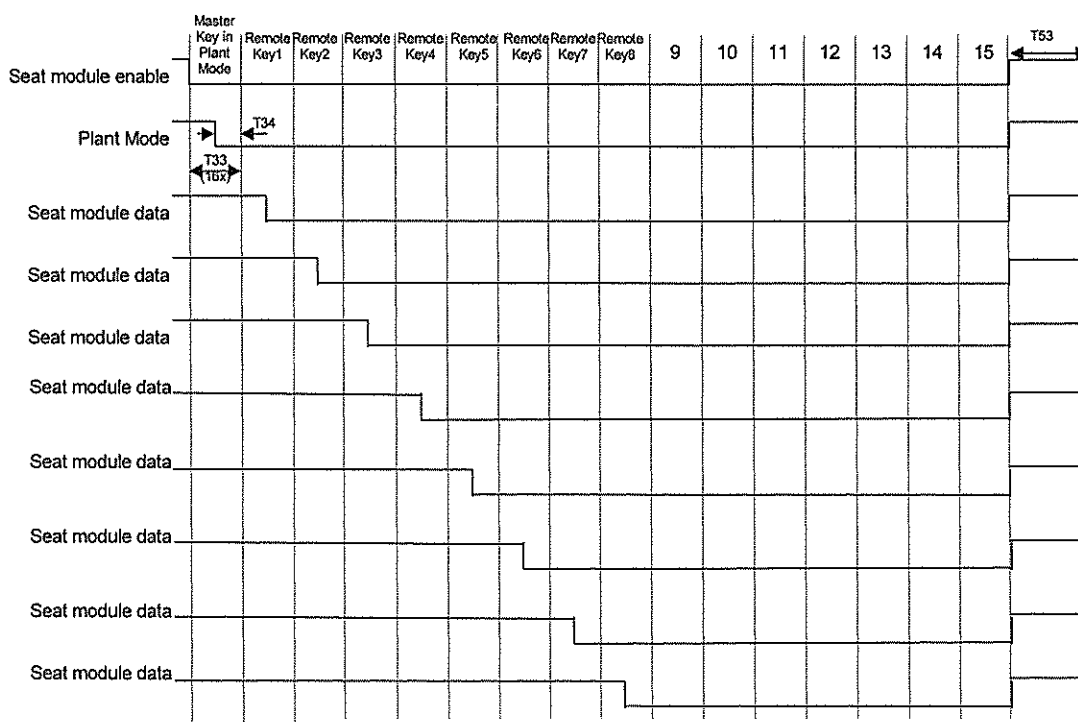
When the driver's door is unlocked using the remote key and ignition is not turned to ignition or start for T54, the BEM shall identify the remote key used and send the relevant information to the Seat Module.

On trained remote key unlock detection, the key pattern (frame) for the memory seat is sent for 10 times as per the diagrams, with high being VOUS and low being VOES. Ignition signalling is performed by driving the seat module enable and seat module data lines in the VOES state when the ignition switch is turned to the ignition or start positions and a delay of T75.

If the vehicle speed is over the 'Memory Seat Speed Limit', T76 or CAN message \$303 from IC is missing or Speed data is invalid, then the seat module data line will be set high (VOUS) for Territory Vehicles only. If the 10 frames for the priority key data are still being transmitted, then they shall have priority over the ignition signalling until they have been completed.

If the vehicle receives another remote key unlock while the 10 frames are still being transmitted, then the current frame shall be completed before a new set of 10 frames is begun.

If no remote keys are trained to the BEM, then plant mode signalling shall occur. If no remote key was pressed before the ignition being turned to ignition or start, then only ignition signalling shall occur after a delay of T75.



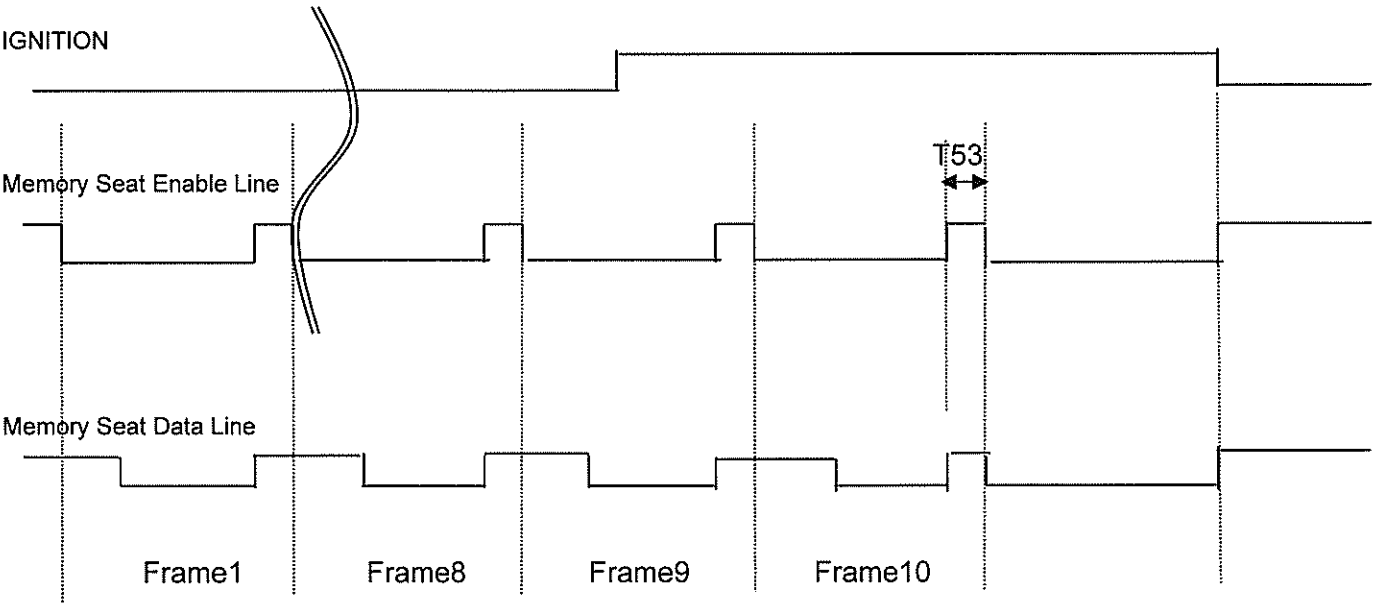


Figure showing Multiple Frames sent – Ignition goes low after 10 frames sent

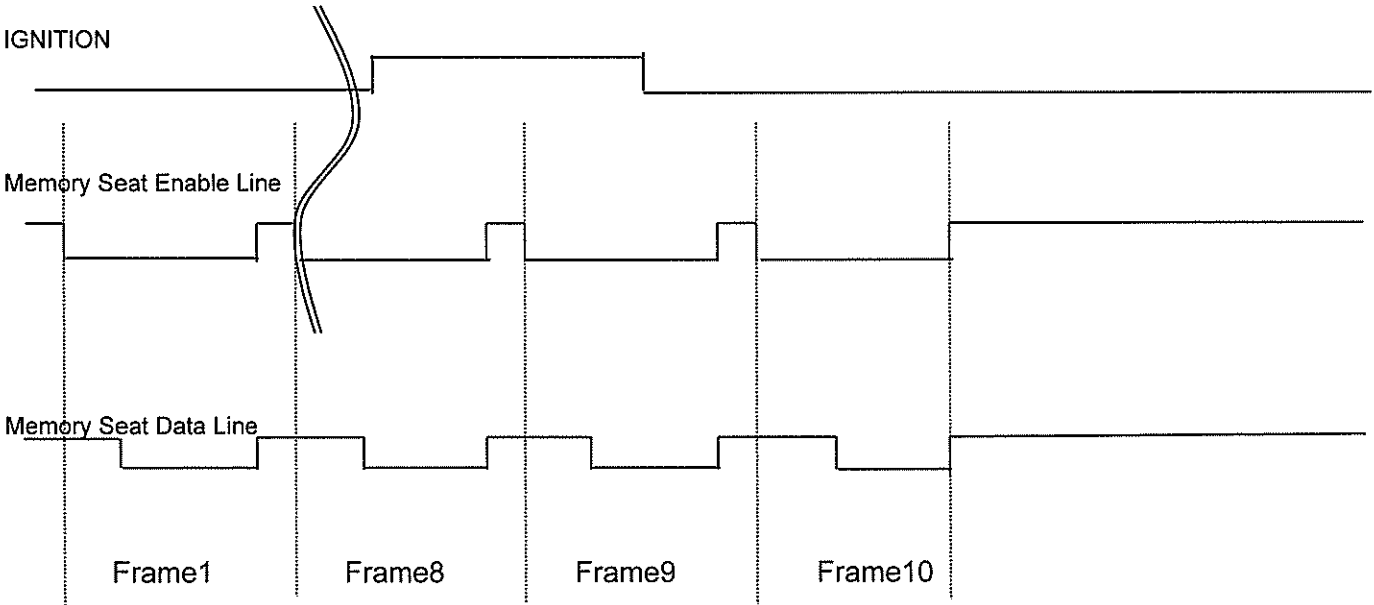


Figure showing Multiple Frames sent – Ignition goes low before 10 frames sent

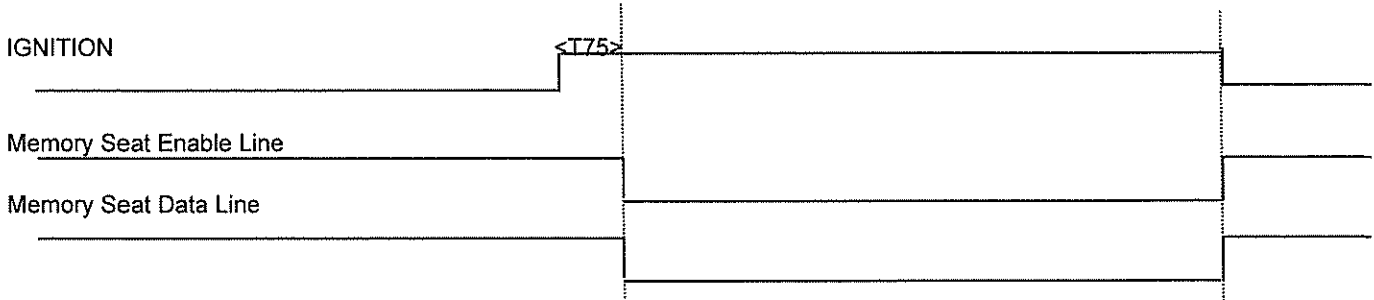


Figure showing signal when no memory seat data is being transmitted.

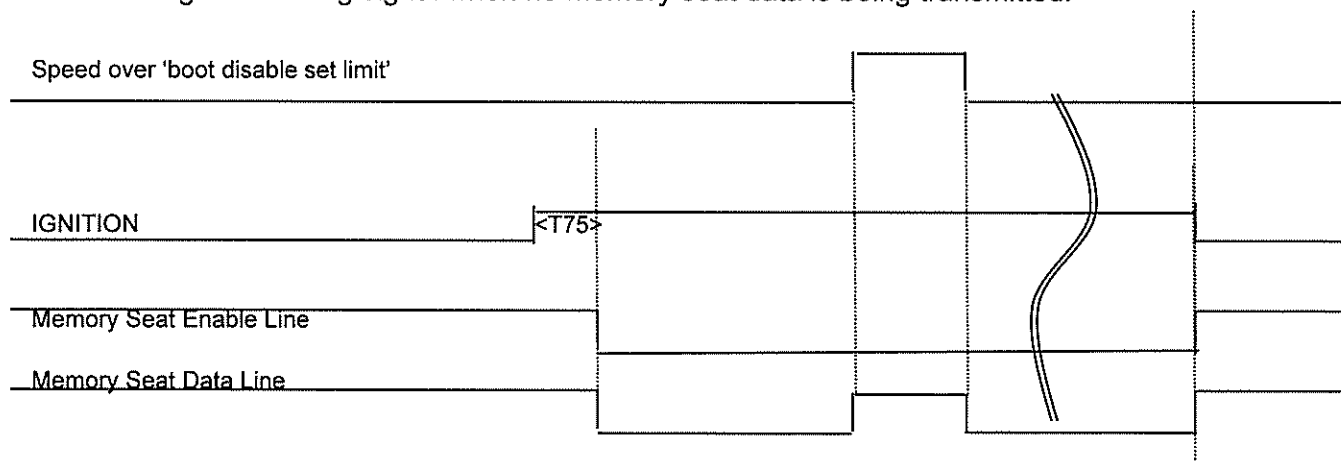


Figure showing signal when no memory seat data is being transmitted and related to vehicle speed.

The BEM sends the CAN message indicating any of the eight trained remote key's unlock button pressed when ignition and accessories off by setting relevant Priority key bit and clearing all other priority key bits. CAN priority bits can also be controlled via diagnostic command \$2F on PID \$E115.

From receipt of a remote unlock command the BEM will send the relevant priority key information at all times the CAN line is active.

From receipt of a remote lock command the BEM will set all priority key bits (1-8) to '0' at all times the CAN line is active.

On trained remote key lock detection, the key pattern (frame) for the memory seat for remote key number 15 is sent for 10 times.

Message \$406 BEM_Interface_1

Signal Name	Bit used	After Reset	Example: After first remote key unlock press on the first trained remote key
Priority key 1	14	0	1
Priority key 2	13	0	0
Priority key 3	12	0	0
Priority key 4	11	0	0
Priority key 5	43	0	0
Priority key 6	42	0	0
Priority key 7	41	0	0
Priority key 8	40	0	0
Master Remote Key/Plant Mode	None	None	None

4.11.2 Signalling in Remote Key Plant Mode

While the vehicle responds to the master remote key, no priority key data shall be sent over CAN. If the ignition switch is turned to the ignition or start positions then memory seat frames corresponding to the master remote key shall be transmitted continuously. Once the ignition switch is turned to accessories or off, then the last frame shall be completed and communication shall stop. No frames shall be transmitted in response to an unlock command from a master remote key.

4.12 Automatic Headlamps

4.12.1 Automatic Headlamp Features

When the ignition key is turned to the ignition or accessories positions, setting the lamp switch to the automatic headlamps position can enable automatic headlamps control. Various factors like twilight sensor status, ignition on/off and driver door status will decide headlamps on/off when lamp switch is at "AUTO" position. If "AUTO" is not selected, the mentioned factors can only affect the BEM headlamp control output, but not the headlamp on/off itself. Table A below summarises the conditions contributing to BEM headlamp output decision

The ambient brightness level is determined by reading the frequency signal from the twilight sensor. To prevent unnecessary activation and deactivation of the headlamps, hysteresis and delay times are used.

The BEM has provision for eight different sets of ambient light thresholds in EEPROM to be set in the RBAU plant and configured by Ford or dealers.

Table A

Case	Twilight sensor status	Ignition or Accessory	Driver Door Ajar	BEM Headlamp relay control output
1	Light	On	N/A	Off
2	Light to dark	On	N/A	Delayed off to on (T35)
3	Dark to light	On	N/A	Delayed on to off (T36)
4	Light	On to off	N/A	Off
5	Dark	On to off	Not triggered	Delayed on to off (T37)
6	Dark	On to off	Triggered	Delayed on to off (T38)
7	Light	Off	N/A	Off
8	Light to dark	Off	N/A	Off
9	Dark	Off (less than T37 from On)	Closed to open	If already on then T38 Delay from on to off
10	Dark	Off	N/A	Off
11	Dark to light	Off	N/A	Off
12	Light	Off to on	N/A	Off

Note: T38 is the delayed turn off time with Headlamps on, ignition turned off and driver door open. It will become 0 seconds when the prime low side driver or backup low side driver fails.

T37 is the delayed turn off time with Headlamps on, ignition turned off and driver door not open. It will become 0 seconds when the prime low side driver or backup low side driver fails.

From the results shown on the table:

- When ignition is on, driver door status does not have any effect to BEM Auto-headlamps output.
- Whether Auto-headlamp is selected on the lamp switch or not, auto-headlamp circuit will function to give a BEM headlamp control. The BEM headlamp control will run the headlamp output when lamp switch is put to the "AUTO" position.

Assumptions:

- Twilight sensor will supply an accurate pulse-width-modulated (PWM) signal with different frequency at different light intensity conditions.
- Twilight sensor will supply accurate PWM signal in no more than 0.5 seconds after supplying power to the sensor.
- Humidity, temperature or EMC does not affect the twilight sensor's performance.

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4.12.1.1 Delay Response periods

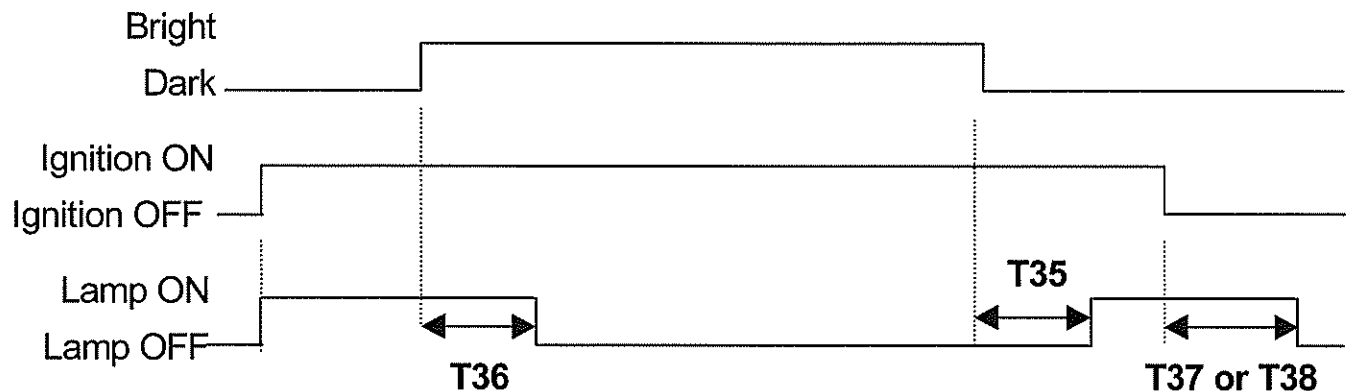


Figure A

There are four delay response periods under different light conditions programmed in the EEPROM. Depending on market requirement and customer expectation, the delay periods can be adjusted accordingly. We shall consider time period **T35**, **T36**, **T38** and **T37**. The delay response period is important in preventing unnecessary flashing of the headlamps.

When the BEM headlamp driver output is enabled, and the ignition is turned to the off position, the BEM headlamp driver output will remain ON for an additional **T38** seconds after the driver door opens. If the driver door remains closed all the time, headlamp driver output will be disabled after **T37** seconds.

When ignition is at the run or accessory position, the BEM headlamp driver output will be enabled/disabled based on ambient light level. The delay response period for the BEM headlamp driver output to react to a bright condition is **T36** seconds, while the delay response period for the BEM headlamp driver output to react to a dark condition is **T35** seconds.

During **T37** and **T38**, when a remote lock signal is received, BEM will turn the headlamp driver outputs off after a period of **T62**.

If delay turned-on time (**T35**) value as read from the EEPROM is outside the positive tolerance, i.e., excessively long, a default failsafe mode will apply. During this mode, the auto-headlamp output will be enabled when ignition key is in the Accessories or Ignition ON position, and will be disabled when ignition key is in the Ignition OFF position.

4.12.1.2 Hardware and Software Control

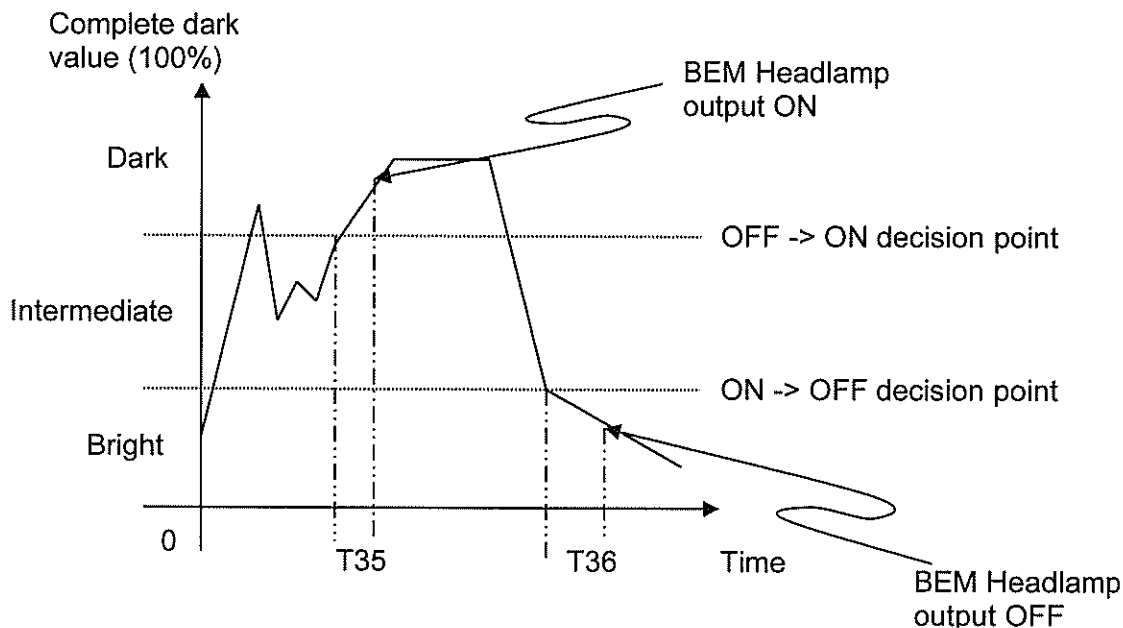


Figure B: Condition: Ignition or Accessories On

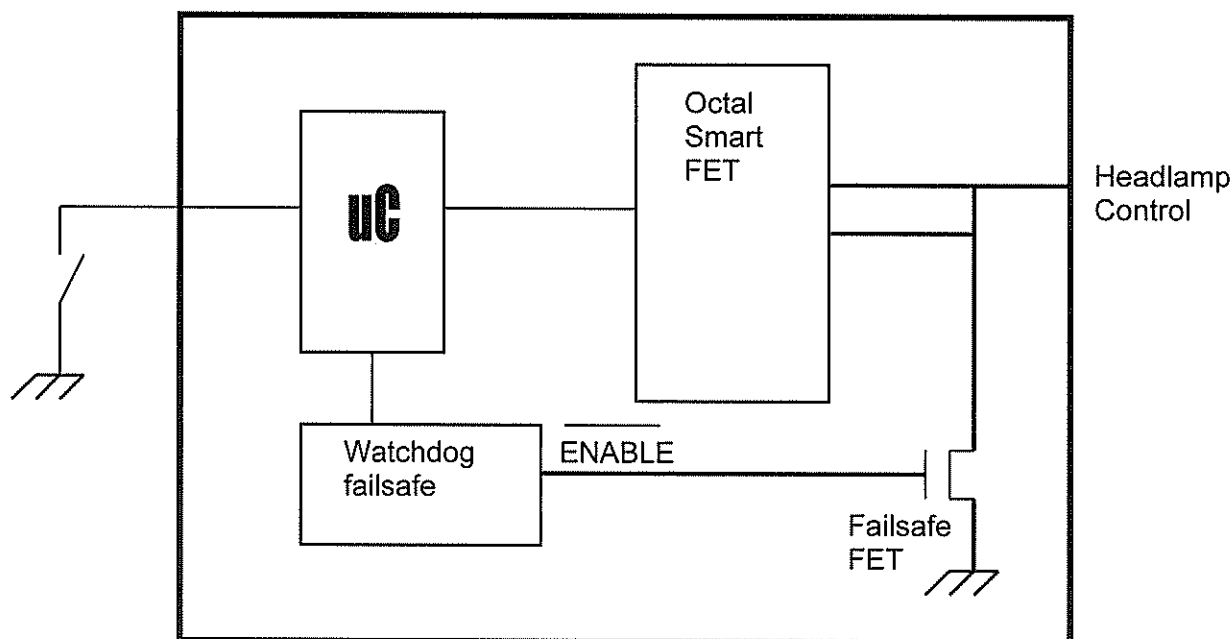


Figure C: BCM Block Diagram

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There are two low side drivers controls automatic headlamp control system. The 2 control signals are an octal smart FET and a failsafe FET. When the ignition switch is at the run or accessories position under a dark condition, the microprocessor will turn on the auto headlamp.

When turning, it will turn Failsafe FET first and do the diagnostics on it, then turn on the Octal Smart FET. If condition is bright and the BEM headlamp output needs to disable active drivers.

Figure B shows the decision points and timing for the BEM control outputs to respond at various conditions. Both outputs perform the same function and are in support of one another, such that the auto-headlamp feature will still be functioning if one of the output drivers fails.

There are 8 EEPROM programmable values for each of OFF to ON and ON to OFF decision points. RBAU manufacturer can control the EEPROM programmed values, while the dealer can only select which of the programmed values to be used. In addition, there is a set of default values that cannot be changed. This set of default values is being applied when the BEM leaves the factory. All On/Off threshold values, as supplied by the customer, will be referred to frequency in Hz obtained from the twilight sensor. Selection from the 8 pre-programmed values will be done by diagnostic commands with diagnostic tools. Table D and E shows the values set.

Each table, D and E, shows the switch on points and the switch off points of the Auto Headlamps defining various settings for dusk to determine when Headlamp relays are to be enabled. Likewise various settings for dawn to determine when the headlamp relays are to be disabled. There is an intermediate section where there is no change in the BEM control output. Eight EEPROM programmable values for each of OFF to ON transitions are selectable through diagnostics (i.e. 0 to 7). A further eight EEPROM programmable values for each of the ON to OFF transitions are also selectable through diagnostics (i.e. 0 to 7). RBAU control the EEPROM programmed values at their manufacturing plant. Only the dealer can select the programmed values to be used. All 16 values will be referred to as a frequency in Hertz, obtained from the twilight sensor.

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Table D: Headlamp On/Off thresholds with Front Wiper OFF:

Settings selection (located in EEPROM)	PID \$7141		Frequency (Hz)	Tolerance (%)
	Byte 1	Byte 2	Decimal	
Headlamp output OFF to ON:				
0x00	0xFC	X	320	±0.5
0x01	0xFD	X	310	±0.5
0x02	0xFE	X	300	±0.5
0x03	0xFF	X	290	±0.5
0x04	0x00	X	280	±0.5
0x05	0x01	X	270	±0.5
0x06	0x02	X	260	±0.5
0x07	0x03	X	250	±0.5
0x08	0x04	X	240	±0.5
Headlamp output ON to OFF:				
0x00	X	0xFC	230	±0.5
0x01	X	0xFD	210	±0.5
0x02	X	0xFE	190	±0.5
0x03	X	0xFF	170	±0.5
0x04	X	0x00	150	±0.5
0x05	X	0x01	130	±0.5
0x06	X	0x02	110	±0.5
0x07	X	0x03	90	±0.5
0x08	X	0x04	70	±0.5

Table E: Headlamp On/Off thresholds with Front Wiper ON:

Settings selection (located in EEPROM)	PID \$7141		Frequency (Hz)	Tolerance (%)
	Byte 1	Byte 2	Decimal	
Headlamp output OFF to ON:				
0x00	0xFC	X	170	±0.5
0x01	0xFD	X	165	±0.5
0x02	0xFE	X	160	±0.5
0x03	0xFF	X	155	±0.5
0x04	0x00	X	150	±0.5
0x05	0x01	X	146	±0.5
0x06	0x02	X	142	±0.5
0x07	0x03	X	138	±0.5
0x08	0x04	X	134	±0.5
Headlamp output ON to OFF:				
0x00	X	0xFC	132	±0.5
0x01	X	0xFD	129	±0.5
0x02	X	0xFE	126	±0.5
0x03	X	0xFF	123	±0.5
0x04	X	0x00	120	±0.5
0x05	X	0x01	117	±0.5
0x06	X	0x02	114	±0.5
0x07	X	0x03	111	±0.5
0x08	X	0x04	108	±0.5

The BEM day/night signal reflects the twilight sensor status defining whether the vehicle is being operated in day or night time ambient light conditions. CAN Message \$403 – IllumMode is used to publish day/night brightness. In failsafe condition, the CAN bit will publish autoheadlamp status (on or off).

Table F: Day/Night Brightness thresholds with Front Wiper OFF:

Settings selection (located in EEPROM)	PID \$7141		Frequency (Hz)	Tolerance (%)
	Byte 3	Byte 4	Decimal	
Day/Night Brightness OFF to ON:				
0x00	0xFC	X	340	±0.5
0x01	0xFD	X	330	±0.5
0x02	0xFE	X	320	±0.5
0x03	0xFF	X	310	±0.5
0x04	0x00	X	300	±0.5
0x05	0x01	X	290	±0.5
0x06	0x02	X	280	±0.5
0x07	0x03	X	270	±0.5
0x08	0x04	X	260	±0.5
Day/Night Brightness ON to OFF:				
0x00	X	0xFC	230	±0.5
0x01	X	0xFD	210	±0.5
0x02	X	0xFE	190	±0.5
0x03	X	0xFF	170	±0.5
0x04	X	0x00	150	±0.5
0x05	X	0x01	130	±0.5
0x06	X	0x02	110	±0.5
0x07	X	0x03	90	±0.5
0x08	X	0x04	70	±0.5

Table G: Day/Night Brightness thresholds with Front Wiper ON:

Settings selection (located in EEPROM)	PID \$7141		Frequency (Hz)	Tolerance (%)
	Byte 3	Byte 4	Decimal	
Day/Night Brightness OFF to ON:				
0x00	0xFC	X	330	±0.5
0x01	0xFD	X	320	±0.5
0x02	0xFE	X	310	±0.5
0x03	0xFF	X	300	±0.5
0x04	0x00	X	290	±0.5
0x05	0x01	X	280	±0.5
0x06	0x02	X	270	±0.5
0x07	0x03	X	260	±0.5
0x08	0x04	X	250	±0.5
Day/Night Brightness ON to OFF:				
0x00	X	0xFC	220	±0.5
0x01	X	0xFD	200	±0.5
0x02	X	0xFE	180	±0.5
0x03	X	0xFF	160	±0.5
0x04	X	0x00	140	±0.5
0x05	X	0x01	120	±0.5
0x06	X	0x02	100	±0.5
0x07	X	0x03	80	±0.5
0x08	X	0x04	60	±0.5

Notes to Table D, E, F and G:

- 1) Tolerance refers to software timing error only.
- 2) Frequencies values shown in table D, E, F and G are reference values as supplied by the twilight sensor supplier. BEM software always works with the input capture values received.
- 3) Factory Setting selections are in *Italic*.
- 4) Table E only provides suggested Headlamp ON/OFF threshold values to Wiper ON situation. These values are subject to change and will be programmed by the Production final tester.
- 5) When a dealer selected a threshold level, the BEM will cross check that the selected OFF to ON threshold is always at a darker condition to the selected ON to OFF threshold. If it is not the case, the BEM will respond to the \$3B block command from the WDS with a negative response code \$22 "condition not correct".
- 6) Setting selections are compared to fixed ROM values 0xFC and 0x04. Setting selection is checked for Out-Of-Range.
- 7) If selected frequency is below 47Hz or above 420Hz, it is considered as an Out-Of-Range case.
- 8) Selected Headlamp ON to OFF threshold is compared with the selected Headlamp OFF to ON threshold. When the selected Headlamp OFF to ON threshold has a higher Input Captured value than that of the selected Headlamp ON to OFF threshold, it is considered as an Out-Of-Range case.
- 9) All Out-Of-Range cases trigger the failsafe feature.

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4.12.1.3 Diagnosis

Diagnosis involves the detection of:

1. Output is short to battery (internal or external fault).
2. Output is short to ground (internal or external fault).
3. Twilight sensor input short circuit
4. Twilight sensor input open circuit

When Auto-headlamps feature is selected while ignition is at the run or accessories position, the microcontroller will turn on Failsafe FET first and check for Error, then will turn on the low side driver on Octal Smart FET. When time to turn off, first Failsafe FET will be turned off and diagnostics done, then Octal Smart FET will be turned off. A frequency counting input and same microcontroller input is used as digital input to detect errors 3 and 4 above on twilight sensor.

With Auto-headlamp feature selected and ignition already at the run or accessories position, auto-headlamp and twilight sensor diagnosis will perform every time the Auto-headlamps are required to switch on and off. The headlamp off diagnostics will not be performed when Ignition switch is at Ignition OFF position.

The twilight sensor diagnostic check will perform before the driver/relay diagnostic check.

4.12.1.3.1 Twilight sensor diagnosis

The twilight sensor circuit feedback will supply a reading under any light condition.

CASE A: Received PWM signal having frequency above the darkest condition. (420 Hz)

Possible error:

- 1) Faulty sensor.
- 2) BEM has internal circuitry failure.

Error message reported: Out of range. (DTC \$9790 set)

CASE B: Received PWM signal having frequency below the brightest conditions (1 to 47 Hz).

Possible errors:

- 1) Faulty sensor.
- 2) BEM has internal circuitry fault.

Error message reported: Out of range. (DTC \$9790 set)

CASE C: No PWM signal is received (0 Hz).

Possible errors:

- 1) External short to battery causes input HIGH.
- 2) External short to ground causes input LOW.
- 3) Sensor has failed open-circuited.
- 4) Sensor has failed short-circuited.
- 5) BEM has internal circuitry fault pulling input HIGH.
- 6) BEM has internal circuitry fault pulling input LOW.
- 7) Sensor is not loaded onto the circuit.

Error message reported: Open circuit or short circuit to ground. (DTC \$9791 set)

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Note: With the above faulty situations (case A, B and C) detected on the twilight sensor portion, as well as an error detected on the sunload sensor portion, BEM output for Auto-headlamps is defaulted ON when ignition key is in the Ignition ON or Accessories position, and output is off when ignition key is in the Ignition OFF position, until the fault is not detected any longer.

4.12.1.3.2 Driver and Relay Diagnosis

When BEM output is turning the Auto-headlamp on.

- Turn on Failsafe FET.
- Use diagnostics feature of Octal Smart FET driver to detect faults.
- Turn on Octal Smart FET.

When BEM output is turning the Auto-headlamp off.

- Turn off Failsafe FET first.
- Use diagnostics feature of Octal Smart FET driver to detect faults.
- Turn off Octal Smart FET.

Right after BEM output turned off the Auto-headlamps, when Ignition switch is not at Ignition OFF position

- Use diagnostics feature of Octal Smart FET driver to detect faults.

With this diagnosis, we can determine if

- Headlamp circuit shorts to a voltage source.

Error message: Auto-headlamp Driver Circuit Failure. (DTC \$9693 set)

- Headlamp circuit shorts to ground.

Error message: Auto-headlamp Driver Circuit Failure. (DTC \$9693 set)

- One of the driver circuits is not performing. (Note 1)

Error message: Auto-headlamp Driver Circuit (DTC \$9693 set)

We cannot test the following in normal operation, as the BEM cannot be sure of whether the "AUTO" position is selected.

- Headlamp relay is loaded or not. (Should be able to test by the micro. controlling the headlamp)
- Lamp switch is loaded or not.
- Broken wiring or connection.

Note 1: One of the drivers is assumed to be working.

4.12.1.3.3 EEPROM settings Diagnosis

To get a meaningful headlamp output ON to OFF, and OFF to ON threshold value, the EEPROM settings must lie within range. Headlamp output will cease to perform and behave as if it is in a failed sensor condition when:

- Any of the two threshold-selection bytes is not between -4 to +4.
- Frequency setting in table is out of range.
- Cross-check that OFF to ON threshold is always at a darker condition to ON to OFF threshold.

Note* Checks are carried out every time an EEPROM settings is called up to be used by the software.

Under these failed conditions, the headlamp output will be defaulted on when ignition key is in the Accessories and Ignition ON position, while the headlamp output will be defaulted off during Ignition OFF position.

4.12.2 Failsafe Feature

Failsafe is a feature that provides Headlights to be ON during Ignition ON, and Headlights OFF during Ignition OFF. BEM will drive the headlamp into Failsafe when BEM has realised a fault in the headlamp driver system. Once the normal operational conditions have been detected again, BEM will get out of the Failsafe mode and provide normal operation.

4.12.2.1 Failsafe Feature1

The 2 drivers design provides control to the headlamp output even if 1 driver fails. The other driver will take over the job and keep control on the headlamp output.

In situation when the microprocessor fails, SBC chip's watchdog feature will turn on the Failsafe driver permanently. In this case, the headlamp needs to be turned off by the switch. Delay response feature will not operate if the microprocessor fails.

4.12.2.2 Failsafe Feature2

Under the situation when the twilight sensor is not loaded and "AUTO" is selected on the lamp switch, the default is having the headlamps on all the time when ignition switch is at the run or accessories position, and the headlamps off when ignition is off.

During a power reset situation where sensor data is not readily available, BEM will assume a dark condition and turn the auto-headlamp output ON, until sufficient data is collected from the sensor to determine the light condition. Upon power reset, the BEM will make sure that sufficient twilight sensor information can be collected within 2 seconds. If the light condition is "BRIGHT", the BEM will respond by setting a delayed turn-off situation to the auto-headlamp output, with a delay time of T36. During cranking, if the BEM should reset (due to low battery voltage), then the auto-headlamp drivers will be turned on Failsafe FET.

Similar situation applies to stop mode. When the car is left untouched for over an hour, the BEM will enter stop mode. If the stop mode is exit by turning ignition key to ignition ON position, BEM will still need the required time (no more than 2 seconds) to collect twilight sensor information. During the data collecting time, the auto-headlamp output will not respond until the new light condition data is received.

Below is a summary of conditions where the failsafe feature applies. With this failsafe feature applied, the auto-headlamp output will be enabled when ignition key is in the Accessories or Ignition ON position, while the auto-headlamp output will be disabled during the Ignition OFF position.

- 1) Microprocessor obtained an out-of-range threshold values from the EEPROM.
- 2) Microprocessor obtained out-of-range threshold selection settings from the EEPROM.
- 3) Microprocessor obtained threshold values from the EEPROM so as to enable auto-headlamp output at a brighter light condition than to disable the auto-headlamp output.
- 4) Faulty twilight sensor. (DTC \$9790 set)
- 5) Faulty sun load sensor. (DTC \$9258 set)
- 6) Twilight/Sun load sensor disconnected. (DTC \$9259, \$9791 set)
- 7) Delayed turn-on time is excessively long.

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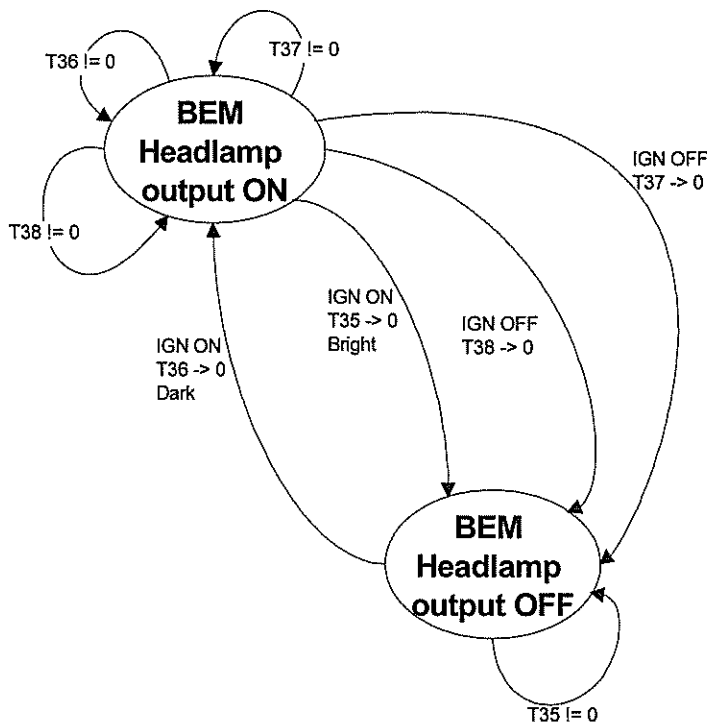
4.12.3 Headlamp Chime Warning

4.12.3.1 Headlamp Chime Warning

When there is a BEM headlamp output failure detected, a Headlamp Chime warning bit will be set for 3 seconds when ignition is just turned from ON to OFF position, and from OFF to ON position. This Headlamp Chime Warning bit is the Headlamp fail warning Buzzer_Request of Message ID \$403. If a failed output is detected, Headlamp Chime Warning bit will still be set until the fault has been fixed. Clear Headlamp Chime Warning bit on ignition switch in 'OFF' position.

4.12.4 State diagram

T35: Delayed BEM headlamp output-enable time.
T36: Delayed BEM headlamp output-disable time, excluding conditions for T37 and T38.
T38: Delayed BEM headlamp output-disable time with output already enabled, ignition just turned off and driver's door triggered.
T37: Delayed BEM headlamp output-disable time with output already enabled, ignition just turned off and driver's door not triggered.



State diagram for auto-headlamp functions
For any discrepancies, the functional description supersedes the above state diagram

4.13 Electronic Immobilisation System

This document complements the Ford design specification "Barra EEC/BEM Immobilisation Code Communications", revision preliminary 2 and should be read in conjunction with this document. The BEM serves as an integral part of the vehicle immobilisation system. The associated components are:

- Ignition key with embedded TIRIS Digital Signature Transponder (DST). The DST ignition key serves as an authority to start and drive the vehicle using traditional mechanical encoding and electronic transponder technology. The DST, TIRIS part number RI-TRP-BRHP, is housed in a 23 mm glass package and utilises challenge response technology. For the BEM to mobilise, the transponder must respond, when requested, with a unique 24-bit identification code, that the BEM has learnt, and a correct 24-bit response to a 40-bit challenge issued by the BEM. The correct response can only be generated when the DST has been taught the 40-bit encryption key by the BEM. Ford must gain concurrence with Bosch regarding the position of the DST within the key head.
- Transceiver Ignition Lock Assembly (TILA). The TILA, in addition to the steering lock and ignition and starter switch functions, serves as an interface between the DST and the BEM through the use of a transceiver circuit. This consists of an electronic circuit and an antenna in a self-contained housing that is attached to the side of the steering lock. Under BEM control, the transceiver uses a RF magnetic field to transmit and receive data with the DST. For further TILA details, refer to Engineering Specification (FOA Barra Body Electronics Module High and Low Series) F005VP0058, section 3.2.
- Body Electronics Module (BEM). The BEM is responsible for verifying that the DST ignition key is one that is recognised as a known DST key, by verifying that its fixed code and challenge response match. Since each BEM uses a unique encryption key, programmed at the factory, a transponder will only be capable of mobilising the vehicle (i.e.: BEM) to which it was last taught. Once this is satisfied, the BEM will mobilise, accept a 32 bit challenge from the PCM module over the CAN bus and send a 24 bit response that is based on an encryption key it has learnt from the PCM, also over the CAN bus.
- Powertrain Control Module (PCM). The PCM is responsible for allowing the vehicle to start, send a challenge to the BEM and, once it has received a correct response to its challenge, then continue running. The PCM should allow at least 2.0 seconds for a correct BEM response before acting to immobilise the engine.

4.13.1 Plant Mode and PCM Security ID Training

An Original Equipment (OE) BEM is set to PCM-BEM, BEM-TILA and BEM-RF Receiver Plant Modes at the factory. At this stage, any attempt to start the vehicle shall result in the BEM attempting to train a DST ignition key. Any immobilisation communication initiated by the PCM shall result in the BEM responding with a 'Ready to Learn Seed' message. Remote entry functionality is restricted, as described in the remote entry software description.

At all times, PCM related PID commands can identify the PCM- BEM plant mode, as well as the PCM-BEM communicating status.

The training sequence in the Ford plant consists of two stages, DST ignition key teaching and learning the PCM security ID.

To teach DST ignition keys to the BEM, each DST key must be placed in the ignition switch and turned to the ignition ON position. For each DST key, the BEM will record the DST's unique identification number and program the DST with its own unique encryption

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key. Code acknowledgement from the BEM will indicate successful teaching of an ignition key. For the first DST taught, there is one acknowledgment, and then there are two for the second DST, etc. In plant mode, a minimum of two and a maximum of eight DST ignition keys may be taught. The acknowledgment of a successfully trained DST ignition key shall occur within time 2.5s of inserting and turning a DST key for training. In the event that programming fails (eg: due to a faulty DST), then no acknowledgment will be produced. Monitoring the error status of the DST ignition key via the diagnostic link can also identify programming failure.

BEM-TILA plant mode (the DST ignition key teaching stage) is exited by the following events:

- Interruption of BEM power once two or more DST ignition keys have been taught.
- Leaving the ignition off for 60 seconds once two or more DST ignition keys have been taught.
- Turning to the ignition position with an already trained DST ignition key once two or more ignition keys have been taught.
- Once eight DST ignition keys have been taught.

If number of DST keys trained is read as more than 8, then assume that BEM is in Plant Mode and number of DST keys trained is 0 (update to eeprom).

Once BEM-TILA plant mode is exited, then the BEM, still in PCM plant mode, is ready to learn the PCM's security ID. If a DST key that is known by the BEM is inserted in the ignition lock and turned to the ignition ON position, then the BEM shall mobilise (due to a successful challenge/response between BEM and TILA) and be ready to accept the security ID from the PCM via CAN. The PCM needs to transmit the 'learn seed' message with the security ID continuously to allow the BEM to read and confirm the data. The BEM will indicate that it has tentatively stored the encryption key by responding with the 'learnt seed' message reflecting the learnt code via the CAN line back to the PCM. The PCM must then acknowledge that the BEM has correctly received the key by sending the 'acknowledge learnt seed' message until the ignition is turned off. Upon receipt of this code, the BEM shall start storage of the security ID within 0.2s and then shall perform a code acknowledgment action after the completion of the storage, and the PCM-BEM plant mode is exited. Under normal conditions, this acknowledgment will begin within time 2.5s after the DST ignition key is turned to the ignition position. Note that in some rare circumstances, other BEM functions may require EEPROM availability, may result in an increase in seed learning time. The BEM shall not accept a challenge from the PCM until the ignition switch has been cycled out of the ignition position. Both PCM-BEM and BEM-TILA plant modes can be exited together when an ignition key with a trained DST is put to ignition ON position to get out of BEM-TILA plant mode. Under this situation, BEM will try learning PCM seed in the same action, and will exit PCM-BEM plant mode, providing that PCM is also in plant mode.

If a BEM has not learnt a security ID and the PCM is not in plant mode, then the BEM will respond with the 'Ready to Learn Seed' message. No acknowledgment will occur in this case. To remedy the situation, the PCM would need to be replaced with one in plant mode, changed back into plant mode or be instructed to perform a parameter reset operation to allow the BEM to learn the security ID.

4.13.2 Normal (P&A) Mode

A parts and accessories (P&A) BEM is set to normal (armed) mode at the factory. A BEM in this state will not allow the plant mode operations to be performed. To install a P&A BEM into a vehicle, both DST ignition key teaching (section 4.13.6, 4.13.7) and PCM parameter reset operations (section 4.13.8) will need to be performed.

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Armed mode and PCM security ID status can be identified by the BEM's mobilisation state and current status of PCM communications PIDs respectively.

4.13.3 Normal Operation – BEM Mobilisation

When the ignition switch is in the off position, the BEM is in the armed state. Turning the DST ignition key to accessories, ignition ON or start positions will cause the BEM to commence a challenge of the ignition key's DST. The challenge created by the BEM is randomly determined by the BEM. The BEM's software ensures the randomness of the challenge, even after reset or power up events. For a new challenge, the BEM must also calculate the expected response for the purposes of verifying the DST's answer. In the event that a new challenge is not ready (such as having to perform two challenges in succession), then the old challenge may be reused. When a challenge is used, it is flagged as being 'stale', and initiation of the calculation of a new challenge is begun immediately. The BEM initially sends a randomly generated 40-bit challenge. The DST will only be able to respond correctly to the challenge if it knows the BEM's encryption key. The BEM also requests the fixed ID code from the transponder, which will identify whether it is one that is known by the BEM.

Successful completion of these two steps will change the BEM's mobilisation status to the mobilised state.

Failure of the challenge will result in the mobilisation state changing to immobilised. An error shall be logged and the transponder ignition key status PID will identify the type of error. If an error is detected, the BEM will continue challenging the transponder until it passes or the ignition switch is turned off, except in an Anti-Scan situation. The number of retries can be configured by Bosch, and is currently set to 'infinite'. To trigger an Anti-Scan situation, BEM must receive a DST response with the right data format but incorrect reply. BEM will then stop challenging in the next 20 seconds. BEM will resume challenging after this 20 seconds period and continue challenging until either

- Mobilised state is achieved
- Ignition switch is turned to Ignition off position
- Another Anti-Scan situation is triggered

4.13.4 Normal Operation – PCM Enabling

Once the ignition switch is in the ignition or start positions, the BEM shall look for a 'pre-challenge' message from the PCM, and respond with a 'request challenge' message. Any challenges received from the PCM shall cause the BEM to respond with a 'not ready, reading transponder' message. Once the BEM has mobilised, it shall request a challenge to the PCM and, wait for the challenge message from the PCM to appear on the CAN bus. The BEM shall then use the security ID, initially learnt from the PCM, to calculate a response and transmit it as a 'response code'. The BEM shall respond in this way to each 'challenge' message received. A 'challenge acknowledge' message from the PCM shall indicate to the BEM that the response has been accepted. Diagnostic interrogation of the BEM may be used to monitor the status of PCM communications.

4.13.5 Challenges from PCM after Normal Start Sequence

In the event that the BEM receives a 'pre-challenge' from the PCM after it has seen a 'challenge acknowledge' message, it will assume a re-challenge situation. The BEM shall request PCM challenge and determine the correct response, but will initially respond with the 'not ready, reading transponder' message. It will simultaneously change to the armed state and challenge the DST inside the ignition key. If the DST ignition key is successfully validated, then the BEM will mobilise again and respond to the PCM's challenge with its 'response code'. An invalid DST will result in immobilisation as in the normal operation case.

4.13.6 Dealer DST Key Teach Mode

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This mode may be used by dealers to train DST ignition keys to a new P&A BEM, or for general DST ignition key erasing and teaching.

NOTE: As a security measure, entering dealer teach mode shall erase the BEM's copy of the security ID from memory. This necessitates that the PCM be instructed to perform a parameter reset following completion of dealer teach mode, for the vehicle to be successfully started.

It is recommended that this procedure be performed with the operator sitting inside the vehicle with the doors closed to correctly distinguish mode and code acknowledgments. Dealer teach mode is entered by setting state = \$11 within \$C125 PID. The ignition switch must be in the off position to enter this mode. Dealer teach mode can only be activated during a time window of T49 just after Ignition is turned to the off position. Entry into dealer teach mode and deletion of the PCM security ID is confirmed by a mode acknowledgment. A BEM in plant mode shall also accept a dealer DST ignition key teach mode request.

To teach DST ignition keys to the BEM, each DST ignition key must be placed in the ignition switch and turned to the ignition position. For each DST ignition key, the BEM will record the DST's unique identification number and program the DST with its own unique encryption key. A sequence of code acknowledges will indicate successful teaching of the current DST ignition key. For the first DST ignition key taught, there is one acknowledgment, then there are two for the second DST ignition key, etc. In dealer teach mode, any number of DST ignition keys up to eight may be taught. The acknowledgment of a successfully trained DST ignition key shall normally occur within time 2.5s of inserting and turning a DST ignition key for training. In the event that programming fails (eg: due to a faulty DST), then no acknowledgment will be produced. Monitoring the error status of the DST ignition key via the diagnostic link can also identify programming failure.

Dealer teach mode is exited by the following events:

- Interruption of BEM power.
- Leaving at ignition off for 10 seconds.
- Turning to the ignition position with an already trained DST ignition key once two or more DST ignition keys have been taught.
- Once eight DST ignition keys have been taught.

Exiting of dealer DST ignition key teach mode is confirmed by another mode acknowledgment.

4.13.7 Customer Spare DST Key Teach Mode

Customer spare DST ignition key teach mode allows the vehicle user to train and delete DST ignition keys on a limited basis. The PCM security ID is not disturbed during this process.

For security reasons, availability of this mode can be set at the Bosch plant, and optionally by the Ford plant and the dealers.

By using a suitable diagnostic tool (eg. WDS), customer spare DST ignition key programming access can be enabled or disabled through a diagnostic command. However, Bosch is also able to control the dealer's access to this command. In summary, Bosch may supply the following configurations:

Customer ignition spare key teach mode enable	Dealer override enable	Outcome
Enabled	Enabled	Customer can train his own DST ignition keys, but dealer can change customer's access rights
Disabled	Enabled	Customer can't train his own DST ignition keys, but dealer can change customer's access
Enabled	Disabled	Customer can train his own DST ignition keys and dealer can not change this
Disabled	Disabled	Customer can't train his own DST ignition keys and dealer can not change this

It is recommended that this procedure be performed with the operator sitting inside the vehicle with the doors closed to correctly distinguish mode and code acknowledgments. To enter customer spare DST ignition key teach mode, the operator must have two DST ignition keys that are already trained to the BEM. One of the DST ignition keys must be inserted into the ignition switch, turned to the accessories position and then removed. Within time 5 seconds, the other trained DST ignition key must be inserted into the ignition switch, and turned to the ignition position. At this point a mode acknowledgment shall occur to confirm entry into the teach mode.

Once teach mode has been entered, the user may teach new DST ignition keys or delete any other DST ignition keys known by the system.

When deletion of DST ignition keys is performed, only the DST ignition keys that are not used in the teach sequence are deleted. The two DST ignition keys that are used in the sequence are not deleted. To delete up to six DST ignition keys, the second DST key must be cycled to the ignition position a second time and then removed, followed by the first DST key, which must also be turned to the ignition position. At this point, the BEM will perform two code acknowledgments to indicate that only two DST ignition keys are now taught (the two that were just used to perform the sequence). A mode acknowledgment will then indicate completion of the teaching mode.

If the user intends to teach new DST ignition keys to the BEM, then after entering teach mode, up to six more DST ignition keys can be inserted into the ignition switch and turned to the ignition position. If teaching is successful, then code acknowledgments corresponding to the total number of DST ignition keys taught will occur.

Customer spare DST ignition key teach mode is exited by the following events:

- Interruption of BEM power.
- Leaving at ignition off for 10 seconds.
- Turning to the ignition position with an already trained DST ignition key, except the second DST key for entering Customer DST teaching mode. The second DST is used for starting the deletion procedure.
- Once eight DST ignition keys have been taught.

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4.13.8 PCM Parameter Reset

A PCM may be instructed to perform a parameter reset on the BEM. This may be used when a BEM or PCM is replaced in the field; the dealer DST ignition key teach mode is performed; or the BEM is otherwise not ready to receive a security ID from a PCM that is in plant mode.

The BEM must first be mobilised by turning the ignition switch to the ignition position with a taught DST. The BEM will then be in a state where it is ready to receive a challenge from a PCM. (In the event that the BEM is still waiting for a security ID from the PCM, this procedure can still be used). The BEM must not have received a challenge response acknowledgment sequence from the PCM on this ignition cycle; otherwise, it will ignore any attempt to do a parameter reset. The ignition switch must be turned off and back to ignition to clear this situation.

The PCM must have been instructed to perform a parameter reset using the appropriate diagnostic tool and commands. Instead of a challenge, the PCM is to send the 'learn seed' message to the BEM. In the event that the BEM has previously learnt a seed (or is a P&A BEM), it shall respond with a 'seed learnt previously' message. If the PCM then sends the correct 'erase seed' message, then the BEM shall then respond with a 'ready to learn seed' message. Once at this state, the remainder of the parameter reset procedure follows the same steps as the plant procedure does.

4.13.9 Transceiver Diagnostics

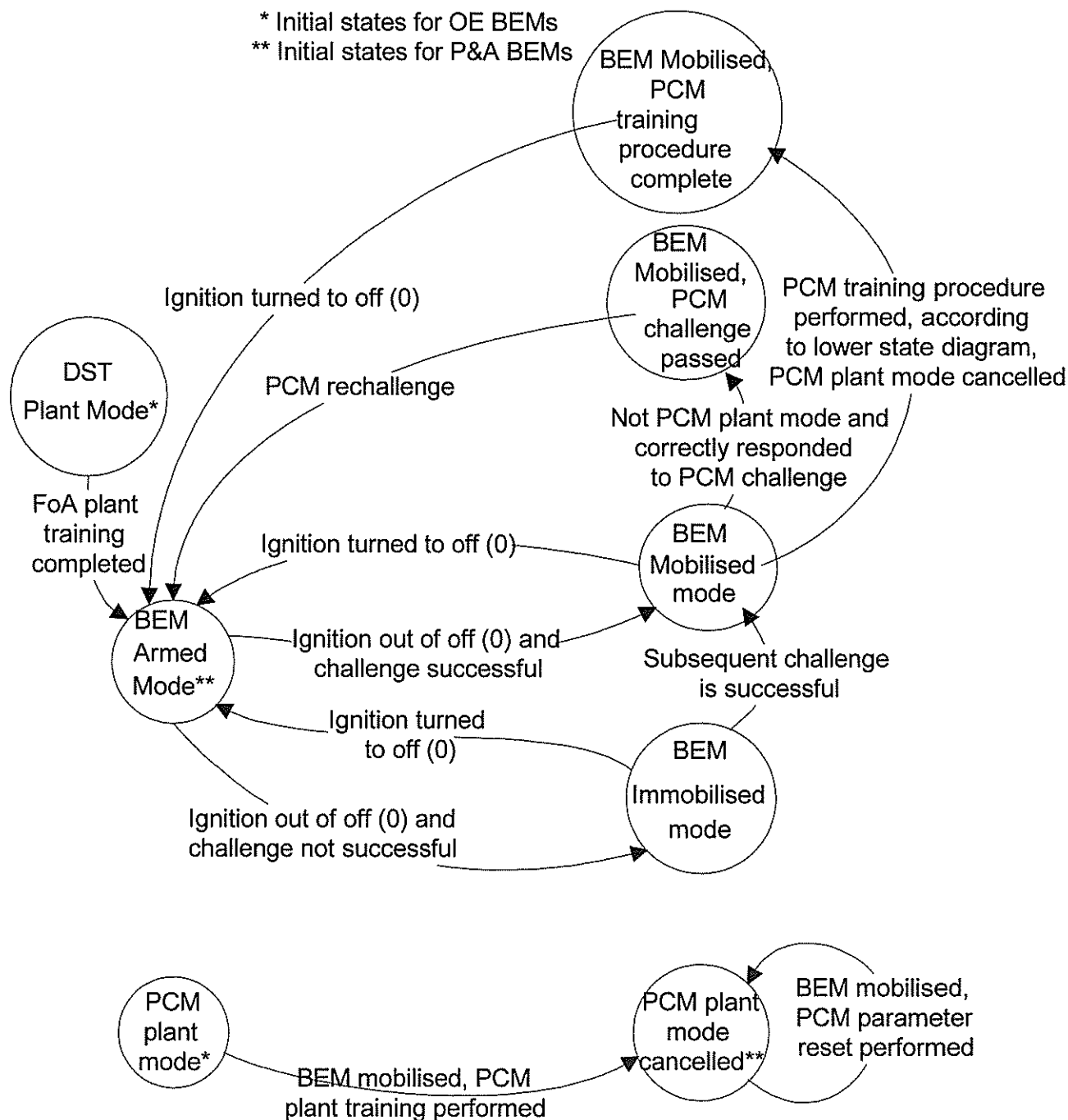
Theoretical analysis shows that the following individual error events may be detected by exercising the TILA transceiver with a DST ignition key inserted.

- BEM transceiver outputs open circuit or short to ground: (DTC \$FFF8)
 - Transceiver supply open circuit.
 - Transceiver transmit data open circuit.
 - Transceiver supply short circuit to ground.
 - Transceiver transmit data short circuit to ground.
- BEM transceiver outputs short to supply: (DTC \$FFF7)
 - Transceiver supply short circuit to supply.
 - Transceiver transmit data short circuit to supply.
- Transceiver internal hardware error (note that interface hardware errors shall fall within one of the above scenarios): (DTC \$A103)
 - Antenna driving circuit.
 - Antenna receiving circuit.
- Less than two DST programmed to vehicle (DTC \$9213)
- Faulty transponder or read only type transponder detected (CRC check fails, suggesting data is not formatted correctly). (DTC \$9602)
- Transponder responds incorrectly to challenge response or DST ID unknown. (DTC \$9601)

BEM Immobilisation System

* Initial states for OE BEMs

** Initial states for P&A BEMs

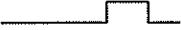
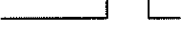

State diagram for BEM immobilisation functions

For any discrepancies, the functional description supersedes the above state diagram

4.14 Code and Mode Acknowledgment Sequences

4.14.1 Code Acknowledgment Sequence

The following sequence is used as a system acknowledgment that an item (DST ignition key or remote key) has been taught successfully. The door locks are toggled into one position and then back again, with an intermediate delay time. Refer to the following table:

Action	Output		Intermediate Time at VOUS	Time at VOES
Condition 1: Lock to unlock to lock	Door Unlock Output		T31	T19
	Door Lock Output		T31	T19
Condition 2: Unlock to lock to unlock	Door Unlock Output		T31	T19
	Door Lock Output		T31	T19

4.14.2 Mode Acknowledgment Sequence

The following sequence will acknowledge entry into or exit from both training modes (DST ignition keys or RF remote keys). The door locks are switched to the opposite state. Refer to the following table:

Action	Output		Time at VOUS	Time at VOES
Condition 1: Lock to unlock	Door Unlock Output		—	T19
Condition 2: Unlock to lock	Door Lock Output		—	T19

4.15 One Touch Start

This function is not finalised yet.....

During the One-Touch-Start, the following functions need to be disabled/de-activated:

-

In the following conditions, BEM assumes that the vehicle is in One-touch-start mode:

- When MS CAN Message \$330 indicates that CrankStatusPCM_MS value is not NoAction (i.e. CrankPending or CrankInProgress or LowVoltageCrankAttempt). Note: CAN Message \$330 should not be detected missing.
- For the time period Txx after ignition key detected as in Start Position (Ignition high and Accessory low).

4.16 General Alarm Description

The ATWS (Anti Theft Warning System) provides an audible and visual alarm at the exterior of the vehicle when an unauthorised access to the vehicle is detected while the system is armed.

Visual alarm, lamps used

The visual alarm consists of the vehicle's hazard lamps.

Types of audible alarm

The audible alarm consists of anti-theft siren (separate from the vehicle horn).

The Orion BEM supports Perimeter Alarm system

Alarm system have four basic states:

- | | |
|-------------|---|
| 1. Disarmed | Any alarm trigger will be ignored at any time. |
| 2. Pre-arm | Transition phase from state disarmed to state armed |
| 3. Armed | Any alarm trigger will lead to alarm activation. |
| 4. Active | Acoustic and visual signal call attention. |

On any change of the alarm system status a CAN message \$406 (BEM_Interface_1) with the corresponding alarm status bits (bit 22-23) is updated.

Alarm Power on Reset Strategy

The system will always store the current alarm state for recovery from power on reset. After power on reset the system will:

- Remain in disarmed state if this was the state before the reset.
- Restart the pre-arming phase if this was the state before the power on reset.
- Trigger an alarm if the system had been in the alarm-active state.
- Return to armed if the system was in the armed state before the power on reset.

Each alarm triggering-input will go immediately to the armed state with out pre-arming if all the following conditions are true:

- Alarm system had been in state armed prior to power disconnection.
- Alarm-triggering input is in its default state (closed or off).

4.16.1 Perimeter Alarm

The perimeter anti-theft feature provides an audible and visual alarm at the exterior of the vehicle when any of the following occur whilst armed:

- an unauthorized access to the vehicle is detected.
- the ignition transitions to RUN or CRANK and Ignition key is not valid (immobilised).

A description of the perimeter anti-theft feature is broken into 3 sections:

- Arming
- Disarming
- Activation

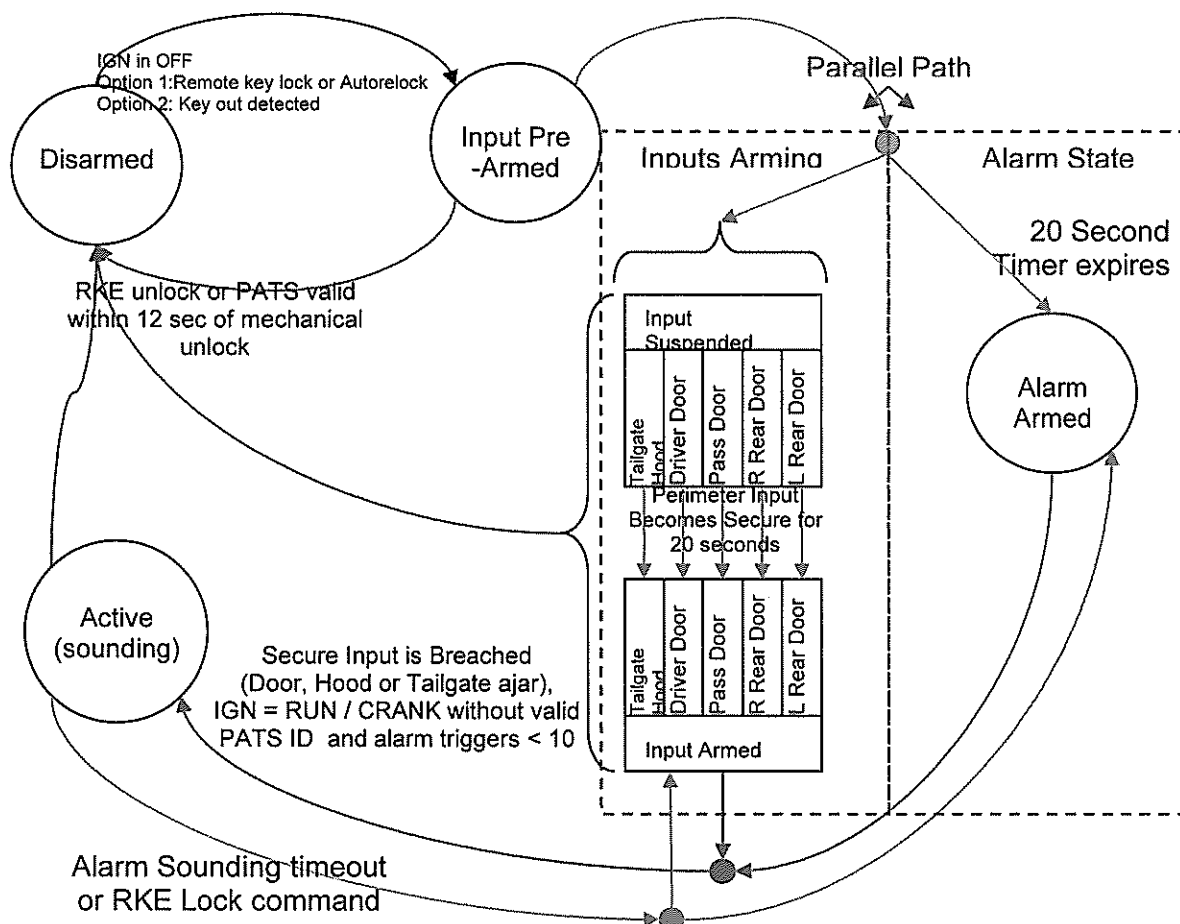


Figure 4.16.A – Alarm States for individual perimeter inputs

Note that "perimeter input" in the above figure refer to Door Ajar, Tailgate and Radio Sense inputs.

4.16.1.1 Arming

Arming activation conditions

The arming of the perimeter anti-theft feature is initiated when either of the following is true:

- If <Alarm Enable on Remote key lock> is enabled, when vehicle has received a remote key lock request including auto-relock while the ignition is in the OFF position.
- If <Alarm Enable on key out> is enabled, when ignition is turned to 'OFF' position and key is removed from the ignition sw before battery saver timeout.

Note: Either <Alarm Enable on Remote key lock> or <Alarm Enable on key out> can be enabled at the same time.

Pre-arm phase

Following the door being closed and arming activated, the PRE-ARM phase is started for each door, boot or hood, which consists of waiting 20 seconds before entering the ARMED state.

Each door, the hood, tailgate has its own individual pre-arm phase with its own pre-arm delay timer. The PRE-ARM phase is a delay period which allows the customer time to open and close any door, hood, or the tailgate without activating the alarm.

After closing the same door, the pre-arm phase for this door starts again with the same pre-arm delay time. Only if the door is left closed for its complete pre-arm phase, it is ready to trigger the

alarm when opened. It is the end of the pre-arm phase, when each input that is in the secure state (e.g., any door is closed) is armed and is ready to trigger the alarm.

Suspended inputs

Any input that is NOT in the secure state (e.g., any door is ajar) enters a suspended mode and will not activate the alarm. Once a suspended input returns to its secure state (e.g., door transitions from Ajar to Closed) and 11 seconds has elapsed, it becomes armed and is ready to trigger the alarm.

4.16.1.2 Disarming**Disarming from Pre-arm state**

During PRE-ARMED state, the perimeter anti-theft feature will DISARM when one of the following events occur:

- The drivers door key cylinder is rotated to the UNLOCK (Mechanical Unlock).
- Remote key Unlock button is pressed.
- Ignition transition to RUN/CRANK with valid PATS read within 750ms (3 attempts).
- Central Unlock button is pressed

Disarming from Armed or Active state

When ARMED, or ACTIVE, the perimeter anti-theft feature will DISARM when one of the following events occur:

- The drivers door key cylinder is rotated to the UNLOCK (Reset) position (given config_alarm_reset_via_key is enabled) and within 12s Ignition transition to RUN/CRANK with valid PATS read within 750ms . A chime on for that duration.
- Remote key Unlock button is pressed.
- Ignition transition to RUN/CRANK with valid PATS read within 750ms (3 attempts).

4.16.1.3 Activation**Alarm activation conditions**

When the perimeter anti-theft feature is Armed, the feature will Activate the alarm when one of the following events occurs without a valid Disarm as described above:

- 0.4s after any door (configurable by DOOR_TRIGGER_TIMEOUT), the hood, or the tail gate becomes ajar, with the following exceptions:
 - The tail gate may become ajar without alarm activation if a remote transmitter trunk release command is received.
 - If the vehicle was mechanically unlocked, (and CONFIG_ALARM_RESET_VIA_KEY is disabled), there is a delay of 11 seconds (started when door becomes ajar) before alarm activation in the event of the driver's front door being opened. A warning chime will be requested from the cluster via the CAN signal. During this time only an ignition transition (with valid PATS is able to disarm the vehicle and stop the alarm activation.

Alarm deactivation to the Armed State

- Remote key panic button press.
- 290 seconds have elapsed since alarm was activated.

Alarm audible warning

When the alarm is activated, the alarm horn will sound for 29 seconds. The alarm tone shall be an intermittent tone: On = 0.5s Off = 0.5s

Visual alarm warning

The visual alarm warning (On = 220ms Off = 220ms) will be active for 290 seconds after which the visual warning will deactivate.

Diagnostics

- The cause of the last 4 alarms shall be stored in an EEPROM for diagnostics purposes.
- Disarming or re-setting of an audible alarm shall not be possible by the diagnostics system except when activated by the diagnostics system.

Alarm retriggering conditions

If the audible alarm has finished there shall be a 5 seconds (configurable via AUDIBLE_HORN_REST) rest time before returning to the ARM (SET) state and any alarm trigger is able to retrigger the alarm.

If the same trigger still exists after the audible alarm time and the rest time, the audible alarm shall retrigger. The visual alarm time will also be restarted.

Maximum alarm triggers

There shall be a maximum total of 10 triggers per arming cycle.

Alarm trigger storage

The last four alarm triggers shall be stored in the BCM.

When the vehicle alarm is not disarmed the BCM shall not allow a transmitter learn.

4.16.3 Visual and Audible Acknowledgements

The following table describes the acknowledgements sent when alarm is in operation.

Type	Acknowledgement	Notes
Disarm due to Remote Unlock	1 Blip/Flash	Flash is part of remote unlock
Disarm due to Remote Unlock	1 Blip/Flash	
Disarm after breach	1 Blip/Flash, Pause, 2 Blip/Flash	First flash is as part of remote key unlock. If PATs is used for disarming, only two flashes and Blip
Arm	2 Blip/Flash	
Arm (Boot Ajar)	3 Blip/Flash	

4.17 Personalisation Interface

4.17.1 Overview

The customer is able to personalize various features within the vehicle to suit their various needs. This specification details those features controlled by the BEM and the interface between the BEM and the ICC.

4.17.2 ICC interface requirements (info only)

- The ICC screen acts as the user interface to show the system status and respond to user inputs.
- The user inputs their configuration by checking, un-checking or changing a slide bar within the GUI of the ICC.
- GUI inputs are speed restricted and the user cannot change settings for speeds >12kph.
- GUI can only be changed after full PATs validation.
- The ICC GUI is auto configurable and constructs itself based on the feature availability and error status from the BEM.
- After power and the system is initialized the ICC stores the current settings from the BEM as a backup. In the event of BEM power reset (resulting in missed messages), the backup values are used to update the display and prevents erratic system operation. Following the BEM reset the display is updated by the 1st BEM message.
- The ICC must receive BEM personalization CAN messages and display the last set status for each feature within 200ms of ignition being turned on (100msec process time within the ICC).
- CAN messages are event-periodic to minimize latency on the bus.

4.17.3 BEM interface requirements

- The BEM controls personalization features associated with Locking, Unlocking, Lighting, Alarm and Secure Boot as detailed in the table below.
- The memory for each feature resides within the BEM and the BEM is master.
- Each personalization feature status requires 3 storage memories, FOB1, FOB2 and mechanical key/ FOB 3-8 users. Where a mechanical key or FOB 3-8 is used the personalization status is recorded in memory 3 so as to not corrupt FOB1 or FOB2 status. The memory 3 location stores the last used vehicle status and is activated whenever FOB1 or FOB2 is not used.
- Personalization feature status is saved on remote key lock.
- The BEM publishes the current FOB user on the CAN bus.
- The BEM must transmit CAN messages showing the last set status for each feature within 100ms of ignition being turned on.
- Default settings for each message are stored in the BEM EEPROM and can be programmed at end of line using CAN diagnostics.
- CAN messages are event-periodic to minimize latency on the bus.

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Table for BEM Personalization

Feature	Setting					
→ Locking						
➤ Drive Away Locking	On	Off	Error	Not available	Inhibited	
➤ Flash Indicators on Lock	On	Off	Error	Not available	Inhibited	
➤ Horn Feedback (One Chirp)	On	Off	Error	Not available	Inhibited	
➤ Reset All						
→ Unlocking						
➤ 2 Stage Unlock	On	Off	Error	Not available	Inhibited	
➤ Flash Indicators on Unlock	On	Off	Error	Not available	Inhibited	
➤ Unlock Puddle On	On	Off	Error	Not available	Inhibited	
➤ Reset All						
→ Lighting						
➤ Auto Headlamp On Sensitivity	-4	-3	-2	-1	0	1 2 3 4
➤ Auto Headlamp Off Sensitivity	-4	-3	-2	-1	0	1 2 3 4
➤ Auto Headlamp Off Delay (At ign Off)	Off		1		2	3 4 (minutes)
➤ Dome Lamp at Ign Off	On	Off	Error	Not available	Inhibited	
➤ Dome Lamp at key out of Ign	On	Off	Error	Not available	Inhibited	
➤ Dome Lamp at Door Ajar	On	Off	Error	Not available	Inhibited	
➤ Dome Lamp at Remote Unlock	On	Off	Error	Not available	Inhibited	
➤ Dome Always Off	On	Off	Error	Not available	Inhibited	
➤ Reset All						
→ Alarm						
➤ Alarm	On	Off	Error	Not available	Inhibited	
➤ Alarm Chirp	On	Off	Error	Not available	Inhibited	
➤ Alarm Option 1 – Remote key	On	Off	Error	Not available	Inhibited	
➤ Alarm Option 2 – Ignition key out	On	Off	Error	Not available	Inhibited	
➤ Reset All						
→ Secure Boot release	On	Off	Error	Not available	Inhibited	
→ Reset All						

Note:

1. Default settings are in bold.
2. If Autoheadlamp on sensitivity and off sensitivity selected such that the table value is not valid, then the BEM will use the closest valid value.
3. Alarm Option 1 and 2 cannot be chosen at the same time. Either one should be selected. Alarm can be disabled only when ignition on and vehicle mobilised.
4. Secure Boot (supported for Orion only) can be disabled only when ignition is on and mobilised. New BEM will have secure boot disabled as default.
5. Lighting Reset will also reset the mood lighting level to 100%.

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4.17.4 Personalisation setting limitations/conditions:

- Secure boot cannot be enabled/disabled for vehicle types Territory and Van.
- Alarm can be enabled/disabled only when vehicle is mobilised. Note: After a BEM reset and alarm is disabled, BEM will check for siren and if found will enable alarm function.
- Alarm configuration can be set to default, only when
 - BEM's Internal flag indicates once siren was detected
 - Vehicle mobilised.
- This will enable alarm and resets all other settings to default.
- For alarm function, only one of <Alarm_Option_1–Remote_key> or <Alarm_Option_2–Ignition_key_out> can be enabled/disabled at once.

4.18 Network Management (MS-CAN)

4.18.1 FNOS Network Management overview

In general the FNOS Network Management of each node can be described by 3 states:

Normal – normal operation of the node; MS-CAN communication required

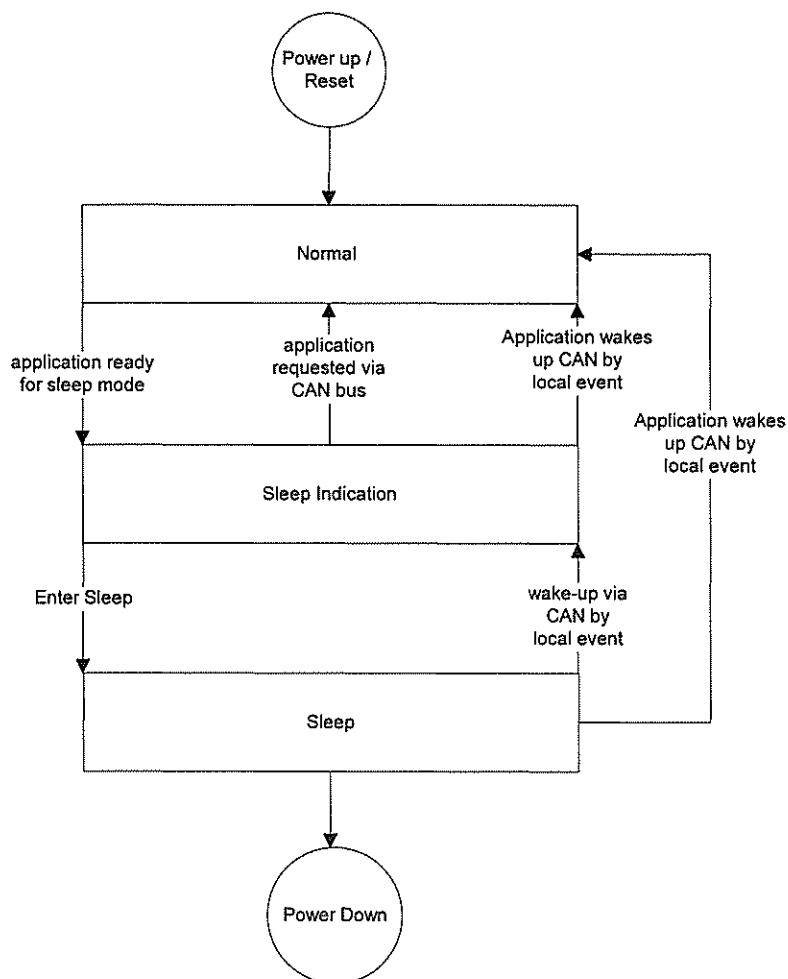
Sleep Indication – normal operation of the node; but no MS-CAN communication required

Sleep – MS-CAN is in sleep mode.

These states are different to the operational states of the node, which means that the node is in normal operation mode while the Network Management is in sleep mode. But if the Network Management is not in sleep mode the node must also be awake.

After a Power up or Reset the node has to enter the normal state for 5 seconds before it could go to the sleep indication state if no normal mode is required.

In figure below the internal behaviour of the Network Management is shown. The transitions between Normal and Sleep indication and the transition from Sleep to Normal are handled by using the Node Management. The Network Management itself handles the other transitions.



Network Management States and Transitions

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4.18.1.1 Wake-up Path

If a node has to send out a message while the MS-CAN is in sleep the following steps must be implemented:

1. Set the Network Management to Normal state.
2. Send out the message.

4.18.1.2 Sleep path

To ensure that a CAN message is really received by all nodes before the MS-CAN enter sleep mode, the sending node must follow the steps below:

1. Hold the Network Management in the Normal state for 1 additional second after sending the signal.
2. After that second it can set the Sleep Indication state.

4.18.1.3 Fault strategy, Busoff and Limphome mode

To guarantee the correct function of the Network Management the following fault conditions have to be taken into account:

1. In the case of a MS-CAN **Busoff mode** the Network Management state Sleep Indication has to be entered. Otherwise the node would keep the bus awake as long as the Busoff mode is active, because the Network Management will start a recovering strategy when Busoff mode is entered due to internal issues of the node, which might lead to a flat battery.
2. In the case that the FNOS enters the **Limphome mode** the Network Management state Sleep Indication has to be entered. If the node does not perform this Network Management state change, it will keep itself alive as long as the Limphome mode is active. If MS_CAN did not enter sleep mode, then after 1s is forced to sleep.
3. If any other fault occurs (e.g. stuck switch input, missing message) a fall back strategy has to be entered which allows the node to enter the NM sleep indication state.

4.18.2 Wake-up Events and Sleep Conditions for BEM

For the test traceability each item is identified by a unique ID, which is build as follows:

NM-node-wscn-number with

node = Node name

wscn = Wake-Up event (W) or Sleep condition (S) for CAN © or node (N)

number = running number for each requirement

The tables specifying the wake-up events and sleep condition have the following columns:

ID = see above

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Signal name = name of the CAN signal

Function = Specification how the signal is generated

Comment = additional information

For Sleep conditions only:

NM User Data (binary) = information about the sleep conditions whether they are set or not; 1 means sleep condition is true, 0 means sleep condition is not true

4.18.2.1 General requirements

NM-ALL-GENERAL-1: Start up initialisation of the network management

After a Power up or Reset the node has to enter the normal state for 5 seconds before it could go to the sleep indication state if no normal mode is required.

NM-ALL-GENERAL-2: Wake-up Path

Put the network management in *Normal* state before sending out the message.

NM-ALL-GENERAL-3: Sleep Path

Hold the bus in the state *Normal* for 1 additional second after sending the signal. After that it can go in Sleep Indication state.

NM-ALL-GENERAL-4: NM control in case of faults

Every node shall guarantee that a fault on any input – e.g. missing CAN message, stuck switch, stuck CAN signal (without missing message) or missing confirmation – does not prevent the bus from going into sleep mode

NM-ALL-GENERAL-5: NM User Data

Every node shall include the information about the sleep conditions (whether they are set or not) within the User Data of the NM message. The following bit coding shall be used: 1 means sleep condition is true, 0 means sleep condition is not true.

Note that several conditions can be set at the same time.

NM-ALL-GENERAL-6: Invalid signal values due to Low/High Voltage conditions

If a node is no longer able to read its Inputs correctly because of a Low or High Voltage condition (<8V respectively >16V), the node shall stop sending application messages but it shall continue sending NM-messages.

Note: This is done by putting the FNOS Interaction layer in the state wait (listen only) or stop (no transmitting and receiving)

The Interaction layer shall be switched on again when the voltage level enters the valid range.

For Orion BEM, the BEM input values will keep the last values, but CAN will be active.

NM-ALL-GENERAL-7: Activated function by CAN signal

If a function has to be executed due to a received CAN signal from another module the receiver module has to keep the bus active as long as the function is active and the MS-CAN is needed for this function.

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4.18.2.2 General Requirements for the BCM:**NM-BCM-GENERAL-1:**

After a remote wake-up the BCM shall be ready to receive message traffic after 20ms (t_ReadyToReceive).

NMS-BCM-GENERAL-2:

After a local wake-up the BCM shall send the application message within (20 – 50) ms after transmitting the first NM message (t_NM-appl).

NMS-BCM-GENERAL-3:

After a local wake-up the BCM shall send the first NM-message latest 70ms after the local wake-up (t_LocalEvent).

4.18.3 Wake-Up Events for MS-CAN by BCM

ID	Signal/Function	Possible CAN bits affected	Event	Comment
NM-GEM-WC-1	RFD Ajar	RFD_Ajar DomeLampActive RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Right front door ajar input change	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-2	RRD Ajar	RRD_Ajar DomeLampActive RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Right rear door ajar input change	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-3	LFD Ajar	LFD_Ajar DomeLampActive RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Left front door ajar input change	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-4	LRD Ajar	LRD_Ajar DomeLampActive RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Left rear door ajar input change	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-5	Tailgate/Tail window Ajar	BootTailgateAjar DomeLampActive	Tailgate/Tail window ajar input change	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-6	Bonnet Ajar	BonnetAjar	Bonnet ajar input change	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-BCM-WC-7	Driver door lock feedback input	RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Change in driver door lock feedback input	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1

ID	Signal/Function	Possible CAN bits affected	Event	Comment
NM-BCM-WC-8	Accessory Switch	AccessoriesSwitch Ignition_Status	Change in Accessorie s switch	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-BCM-WC-9	Remote key lock	Key_1_LockRequest Key_1_UnlockRequest Key_2_LockRequest Key_2_UnlockRequest HazardOnRequest Priority_key_1 Priority_key_2 Priority_key_3 Priority_key_4 Priority_key_5 Priority_key_6 Priority_key_7 Priority_key_8 DomeLampActive RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Valid lock command received	Flasher feedback for remote key lock command. BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-BCM-WC-10	Remote key unlock	Key_1_LockRequest Key_1_UnlockRequest Key_2_LockRequest Key_2_UnlockRequest HazardOnRequest Priority_key_1 Priority_key_2 Priority_key_3 Priority_key_4 Priority_key_5 Priority_key_6 Priority_key_7 Priority_key_8 DomeLampActive RFD_Locked RRD_Locked LFD_Locked LRD_Locked	Valid unlock command received	Flasher feedbacks for remote key unlock command. BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-BCM-WC-11	Remote key boot	HazardOnRequest DomeLampActive	Valid boot command received	Flasher feedback for remote key boot comm BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1 and.
NM-BCM-WC-12	Remote key panic	HazardOnRequest DomeLampActive	Valid panic command received	Flasher feedback for remote key panic command. BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1

Note: After every Wake-up event the BCM keeps the bus awake for at least 5 seconds unless other conditions hold the CAN alive.

4.18.4 Wake-Up Events for BCM by MS-CAN

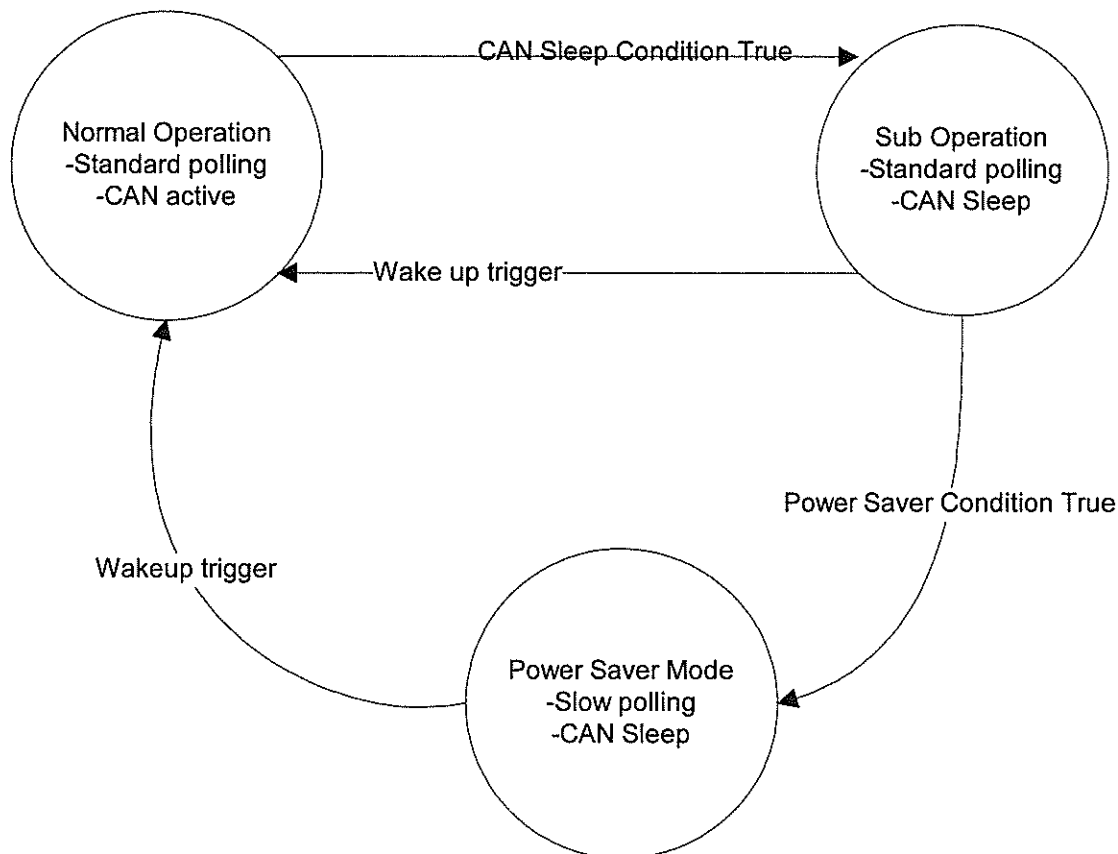
ID	Signal/Function	Possible CAN bits affected	Event	Comment
NM-GEM-WC-1	Msg \$307 - ScreenAuxSwitches_MS IllumBatsaverRequest bit		IllumBatsaverRequest bit on	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-2	Msg \$307 - ScreenAuxSwitches_MS DoorLockSwitch bit	RFD_Locked RRD_Locked LFD_Locked LRD_Locked DomeLampActive	DoorLockSwitch press	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-3	Msg \$307 - ScreenAuxSwitches_MS DoorUnLockSwitch bit	RFD_Locked RRD_Locked LFD_Locked LRD_Locked DomeLampActive	DoorUnLockSwitch press	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1
NM-GEM-WC-4	Msg \$307 - ScreenAuxSwitches_MS DomeLampSwitch bit	DomeLampActive	DomeLampSwitch press	BEM wakes up from sleep and send messages \$403 BEM_Status_MS and \$406 BEM_Interface_1

4.18.5 MS-CAN sleep Conditions for BCM

ID		Bus Sleep entered when	Comment	NM UserData (BINARY)
NM-BCM-SC-1	Ignition	Off		0000 0001
NM-BCM-SC-2	Hazard Warning Sequence	Off	Panic Alarm or Theft Alarm. Hazard flash due remote key press.	0000 0010
NM-BCM-SC-3	Chime Warning	Off	Auto-headlamp	0000 0100
NM-BCM-SC-4	Alarm	Disarmed, prearmed or armed	Including the Thatcham alarm chime request	0000 1000
NM-BCM-SC-5	1 second after door ajar status change and ignition switch in 'OFF' position.			0001 0000
NM-BCM-SC-6	5 second after power up, reset, or entering NORMAL state			0001 0000
NM-BCM-SC-7	Locking/Unlocking relays in-active.	Inactive		0010 0000
NM-BCM-SC-8	Diagnostic mode	Inactive		0100 0000
NM-BCM-SC-9	Internal and External Lights	Dome lamp off Auto headlamp off		1000 0000

4.19 Power Management

The following state diagram describes the BEM states for power management:



The following conditions should be valid to enter power saver mode:

- CAN sleep
- Battery Saver Supply off
- Power Window Supply off
- Ignition Switch in 'OFF' position
- Auto-headlamp off
- Door lock and unlock relays off
- Tailgate and Rear Window relays off
- Demist off
- No eeprom write on progress or pending
- Alarm is armed (if fitted)

4.20 Battery Over Voltage Strategies

When battery voltage is above 18.5 volts, the following functions/outputs will be disabled:

- Demist
- Dome
- Rear Wiper
- Door Locking/Unlocking
- Rear Window and Tailgate Release
- Power Window

4.21 Enabling/Disabling DTC Logging

DTC logging will be disabled during the following conditions:

- BEM detects battery voltage is detected as out of operating range (below 6.5 volts or over 18.5 volts).

DTC logging for missing messages (DTC \$D900) will be disabled during the following conditions:

- BEM detects battery voltage is detected as out of operating range (below 6.5 volts or over 18.5 volts).
- CAN Msg \$330 IC_CombinedPCM_Info – DTC_Logging_MS is read as 0 and CAN Msg \$330 is not missing.

Any DTC should be disabled by BOSCH Plant.

The DTC logging related to following functions can be enabled or disabled by configuring eeprom:

- Memory Seat Control
- Rear Wiper & Wash Control

4.22 VIN Requirements

BEM should have the capability to store and report two instances of VIN (17 character length).

First VIN	Programmed by the End of Line test and configuration system by using Method 2 – writeDataByLocalIdentifier. Once written, it cannot be overwritten. Each subsequent request to write to the VIN block will result in storage of the second VIN. Each subsequent request to write to the VIN block will result in an overwrite of the data stored in the second VIN area. The first VIN will never be cleared or overwritten.
Second VIN	It can only written/overwritten by diagnostic mode 87 using security access.

A diagnostic request to report VIN from the VIN PIDs, or read block \$01 using the ReadDataByLocalIdentifier service, will always result in reporting of the latest VIN stored, or in the instance where there is only a 'first VIN' that VIN will be reported.

4.23 Summary of BEM Functions

Functions	Always	Optional
Heated Backlight (Demist)	X	
Front Wiper and Washer System	X	
Rear Wiper and Washer System		X
Power Window Control	X	
Auto-headlamp Control • Auto Headlamp On Sensitivity • Auto Headlamp Off Sensitivity • Auto Headlamp Off Delay (at ignition off)	X X X X	
Courtesy Lamp (Dome) • Dome Lamp ON at Ign Off • Dome Lamp at key out of Ign • Dome Lamp at Door Ajar • Dome Lamp at remote key unlock • Dome Always Off	X	X X X X X
Battery Saver Supply Control	X	
Mood Lighting	X	
Central Door Locking Functions • Tailgate and Rear Window Release ◦ Secure Boot (Orion Only) • Manual & Remote Central Locking ◦ Flash Indicators on Lock ◦ Horn Feedback (One Chirp) ◦ 2 Stage Unlock ◦ Flash Indicators on Unlock ◦ Automatic Door Re-Lock ◦ Drive-Away Locking	X X	X X X X X X
Electronic Immobilisation System	X	
One-Touch Start	X	
Alarm • Alarm Chirp • Alarm Option 1 – Remote Key • Alarm Option 2 – Arm on ignition key out		X X X X
Personalisation	X	
Diagnostics	X	

• BEM Default Strategies

Message Identifier	Bit	Message Name	Functions Effected	Fault Condition	Comments (BEM Action/Assumption)
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Received by BEM

\$330	32-47	Vehicle Speed	Boot Release	No Message	• 0 km/h, boot can be released
				Invalid Data	• 0 km/h, boot can be released
					• 0 km/h if Ignition is off
			Memory Seat Data and Enable		• No Memory Seat Action i.e. Memory Seat data line is set high.
			Int Wipers	No Message	• 0 km/h, no speed dependent wipers
				Invalid Data	• 0 km/h, no speed dependent wipers
			Drive-Away Locking	No Message	• 0 km/h
				Invalid Message	• 0 km/h
\$307	2	Rear Demist Switch Status	Rear Demist	No Message	• No Demist function
			Remote Key Teach	No Message	• No Remote Key teach
	24	Illumination Battery Saver Request	Battery Saver in Ignition off	No Message	• BEM will assume no requirement to turn on Battery Saver when Ignition off after timeout. This will effect radio reception
	25	Door Lock Switch Status	Door Lock Operation	No Message	• No door lock button function
	29	Door Unlock Switch Status	Door Lock Operation	No Message	• No door unlock button function
	26	Dome Lamp Switch Status	Dome Lamp Control	No Message	• No dome lamp control
\$340	0	Air Bag Deployed	Door Unlock function	No Message	• Doors are left in their current state, no action
\$330	30	DTC Logging	Logging of DTC's	No Message	• DTC's \$D900 not logged.
	16-23	Transmission gear position	Rear Wiper	No Message	• No action, BEM does not perform a Rear Wipe Operation if the Front Wipers are operating.
\$6F6	0-63	PCM Immobilisation transfer	Immobilisation	No Message	• No action, BEM waits for challenge
				Invalid Data	• BEM responds to challenge, up to PCM to verify response and mobilise vehicle. If challenge is incorrect format, BEM does not respond and continues to wait for correct challenge.

Transmitted by BEM

\$403	0	Right front door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	1	Left front door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	2	Right rear door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent

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Message Identifier	Bit	Message Name	Functions Effected	Fault Condition	Comments (BEM Action/Assumption)
	3	Left rear door ajar	Cluster indicates door status	No Input	• Door always closed, 0 sent
	4	Boot/Tailgate ajar	Cluster indicates Boot/Tailgate status	No Input	• Boot/Tailgate always closed, 0 sent
	5	Rear Demist Segment	CDU indicates demist status	No Input	• No Demist jewel, 0 sent
	6	Accessories switch status	Used by other CAN modules.	No Input	• No Accessories, 0 sent
	8	Front Wiper Intermittent switch position	None	No Input	• Out of Intermittent position, 0 sent
	9	Headlamp fail warning	Chime function related to Auto Headlamp driver failure	BEM Feedback not working	• BEM will send 0 as a default.
	10	Hazard on request	Cluster, Remote Entry, Panic, Boot	Remote Functions not working	• Hazards always off, 0 sent
	11	Priority Key 4	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	12	Priority Key 3	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	13	Priority Key 2	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	14	Priority Key 1	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• 1 sent unless reset occurred then 0 sent indicating No memory seat function
	15	Smart Shield jewel on (LED) request	Cluster transponder near lock warning	Transponder in Ignition warning not working	• No warning, 0 sent
	24-31	Sunload Sensor Driver	Climate Control (ACC)	Ignition off, open circuit, short circuit or exceeding 4.7V or less than 0.2V, up to 0.5 sec after ignition on	• \$FF will be sent
	32-39	Sunload Sensor Passenger	Climate Control (ACC)	Ignition off, open circuit, short circuit or exceeding 4.7V or less than 0.2V, up to 0.5 sec after ignition on	• \$FF will be sent
	40	Priority Key 8	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	41	Priority Key 7	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	42	Priority Key 6	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent

Message Identifier	Bit	Message Name	Functions Effected	Fault Condition	Comments (BEM Action/Assumption)
	43	Priority Key 5	Memory Seat, Audio, ACC, mirror	Remote Functions not working	• No memory seat function, 0 sent
	44	Front Wipers On	None	Wipers not working	• No Wiper Operation, 0 sent
	45	Buzzer Request	Buzzer request identifying entry/exit to door trigger disable mode	No audio CAN message received	• BEM will not enter door disable trigger mode, 0 sent
\$6F8	0-63	BEM Immobilisation transfer	Immobilisation	Invalid data from transponder	• BEM responds to PCM with a 'Not Ready, Reading Transponder' message

4.24 Configurable Options

A number of configurable EEprom options are available in the BEM software. These are summarised below:

Function	Factory Setting (Default)	Description	Accessibility
DST Key out dome illumination enable	Enabled	Enables or disables feature of dome lamp illumination with DST ignition key removal from ignition switch.	Bosch plant, Car Display
DST Key off dome illumination enable	Disabled	Enables or disables feature of dome lamp illumination with DST ignition key turning to the off position.	Bosch plant, Car Display
Bosch setting of auto headlamp thresholds	See Auto Headlamp	Bosch programs eight sets of thresholds for use with the auto headlamp system. Each set consists of a high threshold and a low threshold to provide hysteresis.	Bosch plant
Selection of auto headlamp threshold	See Auto Headlamp	Allows one of eight threshold settings to be selected for auto headlamps.	Dealer, Ford, Car Display & Bosch plants
Following functions can be disabled with relevant DTC logging • Seat Memory Module • Rear Wiper • Rear Window Release	Enabled	Allows to enable or disable any combination of functions.	Dealer, Ford & Bosch plants
Taxi Mode	Disabled	Unlocks Passenger doors without unlocking the Drivers Door when using the Central Locking Dash Switch	Dealer, Ford & Bosch plants
Panic Alarm Inhibit	Disabled	Disables/Enables Panic Alarm	Dealer, Ford & Bosch plants
Door Trigger Disable Mode	See Door Trigger Disable Mode	Disables/Enables Door Trigger Disable Mode (alternatively can be manually enabled/disabled via holding the Dome Lamp Dash Switch)	Dealer, Ford & Bosch plants
Hazard Warning Inhibit	Disabled	Disables/Enables the Hazard Lamp Flash	Dealer, Ford & Bosch plants
Horn Warning Inhibit	Disabled	Disables/Enables the Horn Warning function	Dealer, Ford & Bosch plants
Battery Saver Inhibit	Disabled	Disables/Enables Battery Saver	Dealer, Ford & Bosch plants
Courtesy Lamp Inhibit	Disabled	Disables/Enables Courtesy Lamps which if enabled by default enables Battery Saver	Dealer, Ford & Bosch plants
RF Key Unlock Mode	1 stage	Selects between 1 stage or 2 stage Central Door Unlocking (alternatively can be manually selected by pressing both Lock and Unlock buttons on the RF Remote Key)	Bosch plant
Body Type	00	Set as unknown (See Diagnostic Specification Part 2 for other options)	Dealer, Ford & Bosch plants
Over voltage Inhibit	17± 1 Volt	Disables Rear Window and Tailgate Actuator when an approximate Battery Over voltage Level. Programmable from 16 volts. Out of Range is 16V default.	Bosch plant
Front washer motor on	T05	Programmable from 0.5 second to 5.0 second with 0.01 second increment. Out of Range is 1.25 second default.	Bosch plant
Rear Wiper turn off delay after tailgate/window closed	T12	Programmable from 2.5 second to 8 second with 0.01 second increment. Out of Range is 4 second default.	Bosch plant

Function	Factory Setting (Default)	Description	Accessibility
Tailgate/Rear Window Release Time	T18	Programmable from 0.1 second to 1.0 second with 0.01 second increment. Out of Range is 0.8 second default.	Bosch plant
Central Door Unlock Time	T19	Programmable from 0.05 second to 0.5 second with 0.01 second increment. Out of Range is 0.1 second default.	Bosch plant
Delay after tailgate/window shut before restarting the rear wiper motor.	T22	Programmable from 2.5 second to 8 second with 0.01 second increment. Out of Range is 4 second default.	Bosch plant
Automatic central door re-lock time	T29	Programmable from 7 second to 100 second with 0.01 second increment. Out of Range is 40 second default.	Bosch plant
Minimum time needed to press remote key lock and unlock buttons together to change door unlock mode (1-stage, 2-stage)	T30	Programmable from 1 second to 10 second with 0.01 second increment. Out of Range is 3 second default.	Bosch plant
Auto headlamp relay coil delayed turn on time	T35	Programmable from 0 second to 10 second with 1 second increment. Out of Range activates Failsafe condition.	Bosch plant
Auto headlamp relay coil delayed turn off time	T36	Programmable from 0 second to 255 second with 1 second increment.	Dealer, Ford & Bosch plants
Auto headlamp delayed turn off time with Headlamps on, Ign off edge, driver's door closed	T37	Programmable from 0 second to 255 second with 1 second increment.	Dealer, Ford & Bosch plants
Auto headlamp delayed turn off time with Headlamps on, Ign off edge, driver's door open (or open then closed)	T38	Programmable from 0 second to 255 seconds with 1 second increment.	Dealer, Ford & Bosch plants
Maximum time duration between first and second remote key lock to produce horn acknowledgement	T39	Programmable from 1.0 second to 10 second with 0.01 second increment. Out of Range is 3 second default.	Bosch plant
Delayed turn off after receipt of remote lock during the period T37 and T38	T62	Programmable from 0 second to 255 seconds with 1 second increment.	Dealer, Ford & Bosch plants
Rear Wiper Delay after wash motor on	T63	Programmable from 3 second to 7 second with 0.01 second increment. Out of Range is 5 second default.	Bosch plant
Rear Intermittent Wiper Dwell Time	T64	Programmable from 3 second to 6 second with 0.01 second increment. Out of Range is 4 second default.	Bosch plant
Front Wiper Stall Time	T65	Programmable from 2.0 second to 7 second with 0.01 second increment. Out of Range is 4 second default.	Bosch plant
Max Rear Wiper On duration after wash	T67	Programmable from 10 second to 50 second with 0.1 second increment. Out of Range is 30 second default.	Bosch plant
Timeout for tailgate/window button to de-activate	T69	Programmable from 5 second to 90 second with 0.01 second increment. Out of Range is 45 second default.	Bosch plant

Function	Factory Setting (Default)	Description	Accessibility
Rear Tailgate/Window Release Enable Speed	T70	Programmable from 7 km/h to 25 km/h with 1 km/h increment. Out of Range is 12km/h default	Bosch plant
First Hazard Flash On Time for RF Lock	T71	Programmable from 0.05 second to 0.5 second with 0.01 second increment. Out of Range is 0.2 second default.	Bosch plant
First Hazard Flash Off Time for RF Lock	T72	Programmable from 0.05 second to 0.5 second with 0.01 second increment. Out of Range is 0.2 second default.	Bosch plant
Second Hazard Flash On Time for RF Lock	T73	Programmable from 0.05 second to 0.5 second with 0.01 second increment. Out of Range is 0.2 second default.	Bosch plant
First Hazard Flash On Time for RF Unlock	T74	Programmable from 0.05 second to 0.3 second with 0.01 second increment. Out of Range is 0.2 second default.	Bosch plant
Timeout for rear tailgate and window after tailgate or window switch is activated.	T75	Programmable from 0 second to 25.5 second with 0.1 second increment. Default is 20.0 second.	Bosch plant
Speed limit for setting speed Memory Data Line high	T76	Programmable from 1 km/h to 12 km/h with 1 km/h increment. Out of Range is 4km/h default	Bosch plant
Delay after CAN bit for reverse gear is selected before rear wiper operates	T77	Programmable from 0.75 second to 3 second with 0.01 second increment. Default is 1.5 second.	Bosch plant

4.25 Timing Parameters

Parameter	Description [Note * - stored in the EEPROM]	Time (s)	Tolerance	
T01	Heated backlight relay on time	900.0	-1%	+1%
T03	Variable speed dependent intermittent wiper off time	See Sect. 4.3	-2%	+2%
T04	Fixed front intermittent wiper off time	See Sect. 4.3	-2%	+2%
*T05	Front washer motor on time	1.25	-10%	+10%
T06	Front wiper motor delay after washer motor on	0.49	-15%	+15%
T07	Fail-safe wipe after wash wiper motor on time	5.0	-2%	+2%
T08	Minimum intermittent wiper dwell time for wipe on take off	3.2	-2%	+2%
T09	Power window relay on time	3600.0	-1%	+1%
T10	Power window relay on time with door open	40.0	-5%	+8%
*T11	Battery saver time-out (normal mode)	900.0	-1%	+1%
*T12	Battery saver time-out (plant battery saver mode)	180.0	-1%	+1%
T13	Courtesy lamp short fade in and short fade out	1.0	-10%	+10%
T14	Courtesy lamp entry delay or exit delay	20.0	-5%	+5%
T15	Courtesy lamps fade out time following entry delay or exit delay	6.0	-10%	+10%
T16	Courtesy lamp activation on DST key removal activation window	3600.0	-1%	+1%
* T18	Tailgate/Rear Window release time	0.8	-5%	+5%
* T19	Central locking or unlocking time	0.1	-10%	+10%
T20	Hazard flash off time	0.4	-5%	+5%
*T21	Hazard lamp activation time for boot release	1.0	-5%	+5%
* T22	Delay after tailgate/window shut before restarting the rear wiper motor.	4.0	-5%	+5%
T23	Hazard lamp time during horn warning	0.3	-10%	+10%
T24	Horn warning on time	0.02	-10%	+10%
T25	Horn warning off time	0.09	-10%	+10%
T26	Panic alarm duration	30.0	-2%	+2%
T27	Panic alarm pulse width	0.2	-10%	+10%
T29	Automatic central door re-lock time	40.0	-2%	+2%
T30	Remote key mode change time	3.0	-2%	+2%
T31	Code acknowledgment intermediate delay	0.3	-10%	+10%
T32	Remote Key Teach Mode Access Time Window	5.0	-2%	+2%
T33	Time slice used for BEM to Seat Module communication	0.04	-20%	+20%
T34				
* T35	Auto headlamp relay coil delayed turn on time	2.0	-2%	+2%
* T36	Auto headlamp relay coil delayed turn off time	15.0	-2%	+2%
* T37	Auto headlamp delayed turn off time with Headlamps on, Ign off edge, driver's door closed	120.0	-1%	+1%
T37	Auto headlamp delayed turn off time with Headlamps on, Ign off edge, driver's door closed with DC headlamp control driver failure with AC headlamp control driver operating.	0.0	Not applicable	
* T38	Auto headlamp delayed turn off time with Headlamps on, Ign off edge, driver's door open (or open then closed)	60.0	-2%	+2%
T38	Auto headlamp delayed turn off time with Headlamps on, Ign off edge, driver's door open (or open then closed) with DC headlamp control driver failure with AC headlamp control driver operating.	0.0	Not applicable	
* T39	Second press Warning delay	3.0	-2%	+2%
T40	BEM DST ignition key training time	1.0	-50%	+150%
T41	Plant mode maximum time between DST key teaches	60.0	-2%	+2%
T42	BEM security ID storage time after PCM acknowledgment	0.2	-75%	+0%
T43	BEM PCM training code acknowledgment activation delay	0.5	-5%	+5%
T44	Total PCM BEM training time	2.5	-50%	+150%
T45	Dealer DST ignition key teach mode max time between DST key teaches	10.0	-2%	+2%
T46	Customer DST ignition key teach mode entry time window	5.0	-2%	+2%
T47	Customer DST ign key teach mode max time between DST key teaches	10.0	-2%	+2%
T48	Ignition on to PCM challenge response completion (maximum)	1.5	-5%	+5%
T51	Power Down Mode Entry Delay	2.0	-100%	+0%

T52	Remote Entry Receiver Passive Resynchronisation time	30.0	-2%	+2%
T53	Minimum Inter-frame period	0.20	-10%	+10%
T54	Delay from RK unlock to send data frame	0.20	-10%	+10%
T61	Delay headlamp driver outputs off time, during T59 period, counted after a remote door lock is sensed	0.0	Not applicable	
* T62	Delayed turn off after receipt of remote lock during the period T37 and T38	0.0	Not applicable	
* T63	Rear Wiper Motor Delay After Washer Motor On	1.0	-5%	+5%
* T64	Rear wiper Intermittent wiper dwell time	4.0	-5%	+5%
* T65	Time out used for Front Wiper Park Switch position detection	4.0	-2%	+2%
* T67	Maximum Rear Wiper Motor (failsafe) activation time as result of wash function	30.0	-5%	+5%
* T69	Timeout for rear tailgate and window	45.0	-1%	+1%
* T70	Speed limit for disabling rear tailgate and window	12.0 km/h	Not applicable	
* T71	First Hazard Flash On Time for RF Lock	0.2	-5%	+5%
* T72	First Hazard Flash Off Time for RF Lock	0.2	-5%	+5%
* T73	Second Hazard Flash ON Time for RF Lock	0.3	-5%	+5%
* T74	First Hazard Flash On Time for RF Unlock	0.2	-5%	+5%
T75	Tailgate & Rear Window Timeout after Tailgate or Rear Window Switch is activated	20.0	-5%	+5%
* T76	Speed limit for setting speed Memory Data Line high	4.0 km/h	Not applicable	
* T77	Delay after CAN bit for reverse gear is selected before rear wiper operates	1.5	-5%	+5%
T78	Hazard lamp off time prior to horn warning	0.8	-10%	+10%
T79	Time duration for pressing dome dash switch to enter mood lighting	2.0	-5%	+5%
T80	Time duaration for one-touch start	2.0	-5%	+5%
T81	Time delay after wipe-wash for drip wipe to operate	3.0	-5%	+5%
* T82	Mood Lighting ramp up rate (1% per ms)	0.07	-10%	+10%